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A Landmark-Anchored Timetable-Graph Model of Public Transport: Verified Computation of Accessibility, Its Inequality, and the Fare Matrix from Open Data, with Agent-Reported Hospital Bed Capacity for Ambulance and Patient Routing

Research Category

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ABSTRACT

Verified Computation of Accessibility and Its Inequality from Open Data, with Agent-Reported Hospital Bed Capacity for Ambulance and Patient Routing

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Jeepneys, tricycles, minibuses, and UV vans carry most travelers in Albay, Bicol Region (Region V), yet none of their routes, stops, headways, or fares exist in machine-readable form. Nobody can therefore say how many essential places a resident reaches in a given time or what the trip costs, and an ambulance crew has no quick way to find a hospital with an empty bed. This study asked whether a single laptop working only from open data could compute landmark accessibility, its inequality, and the fare matrix, and could also send ambulances and patients to hospitals with free beds. Roads from OpenStreetMap and estimated headways were combined into a landmark-anchored timetable graph, and earliest-arrival times on that graph gave the landmarks reachable within 30 minutes. Inequality in these counts was summarized with the Lorenz curve, the Palma ratio, a population-weighted Gini coefficient, and a Theil index split by municipality. The same model priced trips between landmarks and located the earliest-reachable hospital with free beds. Twelve correctness results (nine propositions and three corollaries) were proved and then checked on synthetic data. Every earliest-arrival query agreed with exhaustive enumeration; the Gini coefficient was scale-invariant; the Theil decomposition held to machine precision; no larger vehicle gained access in any of 11,620 origin–destination pairs; and all 1,000 fare splits were subadditive. In a mock mail network, the e-mail agents that ask hospitals for bed counts rejected all 200 forged replies, and hospital routing agreed with the reference solution in all 5,200 queries. Open data are therefore enough to compute verified accessibility, its inequality, and the fare matrix, and to route patients by available beds, in support of SDGs 3, 10, and 11. The study recommends that local governments and hospitals adopt the model for transport planning, fare monitoring, and ambulance dispatch, and that it be extended to the rest of the Bicol Region.

Keywords: timetable-graph model, earliest arrival time, Palma ratio, Theil decomposition, fare subadditivity

1. INTRODUCTION

When the United Nations General Assembly approved the 2030 Agenda in 2015, it set out 17 Sustainable Development Goals (SDGs) and committed every country to "leave no one behind" (United Nations General Assembly [UNGA], 2015). Two of its targets deal with the ordinary trips this study is concerned with. Target 11.2 asks for transport that is safe, affordable, and accessible for all, and target 3.8 asks for health care that people can use without falling into financial hardship. Both remain far from met. Across the 412 cities that reported data, only about 61.5 percent of urban residents could conveniently reach public transport in 2025 (United Nations Statistics Division [UNSD], 2026). In 2023, about 4.6 billion people worldwide still lacked access to essential health services, and in 2022 some 2.1 billion faced financial hardship because of what they paid out of pocket for care (World Health Organization [WHO] & World Bank, 2025).

Progress on these targets is hard to track where travel depends on informal transport, because jeepney, minibus, and tricycle routes rarely appear in official data (Klopp & Cavoli, 2019). Albay is a case in point. Most residents ride vehicles that keep no fixed route or timetable, and no open record shows how those vehicles link people to markets, schools, and hospitals.

This study does not try to judge how far the province has come on the SDGs. Its aim is narrower: to show that this kind of spatial access can be defined, computed, and verified from open data before anyone goes into the field for a survey. The framework then carries the same approach over to a second problem, directing ambulances and patients to hospitals that still have room. Every result reported here comes either from a mathematical proof or from a synthetic test environment (Section 3.0), and the hospital routing module has not yet been tried in an actual hospital or ambulance service. Section 1.5 explains how the work connects with the SDGs and states what it does and does not claim.

1.1 Research Questions

The study asks one primary question and six subsidiary ones. Questions Q1 to Q4 concern transport; Q5 and Q6 come from the extension to hospital routing.

Primary Question. Suppose that Albay's informal public transport is described by a machine-readable model built only from public sources. Starting from a set of anchor points, how many essential places can be reached within a set travel time, counted from door to door, and how evenly is that number spread across anchors and across municipalities?

Q1. Where no timetable exists, how can departure times be estimated from service headways, and can a time-expanded timetable graph answer earliest-arrival queries for every landmark both correctly and efficiently?

Q2. How does the set of reachable destinations change when roads are filtered by the width, vertical clearance, gross mass, and axle load of the vehicle, and can the vehicle-specific networks be organized so that routing stays practical across many vehicle types?

Q3. Under the regulated fare assumptions of the study, can trips still be routed by fare, arrival time, and number of transfers at a reasonable computational cost?

Q4. Do the chosen inequality measures keep their expected properties when accessibility is computed over a register of landmarks instead of a full partition of the population? In particular, can the Theil index still separate inequality within municipalities from inequality between them?

Hospital Extension. The extension uses the same timetable graph to route ambulances and patients to hospitals. It performs no clinical assessment; the patient's acuity is supplied as an input.

Q5. Can lightweight software agents collect hospital bed-status reports over ordinary e-mail while keeping message volume low, keeping reports recent, and discarding replies that are invalid or that break the study's rules?

Q6. When patients are matched to hospitals by acuity, required service level, reported bed status, and payment class, does an earliest-arrival search find the nearest suitable facility under the model's assumptions, and how do its recommendations change as the patient's need becomes more urgent?

1.2 Review of Related Literature

1.2.1 Measuring Accessibility

In transport research, accessibility describes how easily people can reach opportunities such as jobs, schools, markets, and health services. Hansen (1959) gave one of the first formal measures, in which an opportunity counts for less the more time or money it takes to reach. Much later, Geurs and van Wee (2004) grouped the measures then in use into four families, based respectively on infrastructure, location, persons, and utility. The cumulative opportunity measure of Wachs and Kumagai (1973) belongs to the location family: it simply counts the opportunities lying within a fixed travel-time budget. For equity work its appeal is practical. It has no behavioral parameter to calibrate, and its result can be explained to a non-specialist in a single sentence (Geurs & van Wee,

2004). Levinson and Wu (2020) take a stronger position and argue that a transport system should be judged by the access it provides more than by mobility or congestion.

A cumulative opportunity measure is the main indicator of this study: the number of relevant places that can be reached within a fixed time. A figure such as "how many essential places lie within 30 minutes" is easy to read, and, unlike a behavioral model, the measure needs no demand function and no estimated preferences. A hard cutoff, however, gives no credit to a destination lying just past the threshold. For that reason the study also reports a gravity-style measure as a robustness check, so that such destinations still count for something.

1.2.2 Equity in the Distribution of Access

A provincial average can hide sharp differences and leave some communities poorly served even when the overall figure looks acceptable. Lucas et al. (2016) set out a formal method for judging whether access is distributed fairly, and Pereira et al. (2017) tied the distribution of transport benefits to theories of distributive justice. This line of work brought into transport research several measures first built for income inequality: the Lorenz (1905) curve, the Gini (1912) coefficient, the entropy index of Theil (1967), and the Palma ratio of Cobham and Sumner (2013). Delbosc and Currie (2011), for example, used Lorenz curves and Gini coefficients to study how public transport supply is spread across Melbourne.

1.2.3 Open Data on Informal Transport

Most transit data in circulation follow the General Transit Feed Specification (GTFS). TriMet, the transit agency of Portland, Oregon, created the format with Google in 2006, and it is now maintained by the nonprofit MobilityData (MobilityData, n.d.-a). Digital Matatus was one of the first projects to apply GTFS to an informal system. Williams et al. (2015) describe how university students in Nairobi rode the matatu minibuses with their phones, recorded the routes, and released the results as a feed that journey-planning apps and planners soon took up. Similar mapping followed in other cities. Saddier et al. (2016) used smartphones to map 315 trotro routes in Accra, and the minibus routes of Maputo were mapped on the Nairobi model (Klopp & Cavoli, 2019). In Greater Cairo, Transport for Cairo worked with the Digital Matatus team to map the formal and informal services of six new desert cities and released the data openly (Transport for Cairo, n.d.). Nearer to Albay, the former Department of Transportation and Communications released GTFS data for Metro Manila covering rail, bus, and jeepney routes, and trip planners such as Sakay.ph have drawn on those data since the Philippine Transit App Challenge of 2013 (Narboneta & Teknomo, 2015). Again and again, the dataset lasted longer than the first analyses built on it.

Two lessons follow for a school research team. The first, often noted, is that the feed tends to be the most lasting product of such a project. The second is that every one of these projects started with field mapping, and field mapping is exactly the step that fails when the team is three students, the schedule is one school term, and some routes climb into upland barangays where a student should not travel alone. The team therefore reversed the usual order and asked, before any field campaign, how much could be defined, proved, and computed.

1.2.4 Journey-Planning Algorithms for Timetable Networks

Two standard models turn a timetable into a graph. The time-expanded model gives every departure and every arrival its own node, so that finding the best journey becomes a shortest-path problem on a fixed directed graph. The time-dependent model keeps one node per stop and lets arc travel times depend on the clock. When no arc weight is negative, Dijkstra's (1959) algorithm works in either model; Cormen et al. (2022) prove its correctness and give its running time with a binary heap. Larger feeds call for faster methods. The Connection Scan Algorithm of Dibbelt et al. (2018) and RAPTOR (Delling et al., 2015) both do without a priority queue, but they place stricter demands on how transfers and footpaths are modeled.

1.2.5 Landmarks as Units of Analysis

Any analysis by administrative zone runs into the modifiable areal unit problem: results depend on how large the zones are and where their borders fall (Openshaw, 1984). In Albay, the 720 barangays (Philippine Statistics Authority [PSA], n.d.) differ greatly in size. A barangay in the poblacion of Legazpi City may measure only a few hundred meters across, while an upland barangay is far too large to be summarized by a single point.

The measure used in this paper avoids zones altogether. Its index set is a finite list of actual places, each listed by name in at least one open register (Table 2). Typical members are a public transport terminal, a senior high school, a public market, a district hospital, and a municipal hall.

1.2.6 Philippine and Albay Setting

In the Philippines, part of the responsibility for planning routes now rests with local governments. The Department of Transportation's Omnibus Franchising Guidelines (DOTr, 2017), which underpin the Public Utility Vehicle Modernization Program, direct every local government unit (LGU) to prepare a Local Public Transport Route Plan with evidence behind each proposed route, and the Department of the Interior and Local Government issued the guidelines that LGUs follow in preparing it (DILG, 2018). Fares, however, are set nationally by the Land Transportation Franchising and Regulatory Board (LTFRB). The latest increase was scheduled for March 19,

2026, but the President suspended it that same day (Cabalza, 2026), and it took effect only on September 28, 2026 (Daguno-Bersamina, 2026; Nieto, 2026). Since then, a traditional jeepney has charged ₱14.00 for the first 4 km and ₱2.00 for each additional kilometer. A modern jeepney charges ₱17.00 for the same 4 km and ₱2.40 for every added kilometer, and an ordinary provincial bus charges ₱12.00 for the first 5 km and ₱2.20 per kilometer beyond that (Daguno-Bersamina, 2026; Nieto, 2026). Separate laws grant a 20 percent fare discount to senior citizens (Republic Act No. 9994, 2010), persons with disabilities (Republic Act No. 10754, 2016), and students (Republic Act No. 11314, 2019). These discounts write into fare policy the concern for older riders, riders with disabilities, and other at-risk passengers that SDG target 11.2 calls for (UNGA, 2015). The province has 18 LGUs and 720 barangays (PSA, n.d.), a land area of about 2,575 km² (PhilAtlas, n.d.), and 1,374,768 residents as of the 2020 census (PSA, 2021).

1.2.7 Emergency Care, Hospital Levels, and Bed Reporting

Operations research on emergency medical services has concentrated on the ambulance, mainly on where to station units and how to move them as calls come in (Brotcorne et al., 2003). The hospital at the other end of the trip has drawn less attention, and it is the focus here. Three established practices govern the choice of hospital.

The first is triage. The Emergency Severity Index sorts an arriving patient into one of five levels, from those who need immediate life-saving intervention to those expected to use no hospital resources at all (Gilboy et al., 2011). The second is licensing. The Department of Health (DOH) classifies hospitals by service capability (DOH, 2012). A Level 1 hospital must have, among other things, Level 1 imaging, a secondary clinical laboratory, isolation and maternity facilities, and an operating room with a recovery room. Level 2 adds a general intensive care unit (ICU), a neonatal ICU, and facilities for high-risk pregnancy, and Level 3 adds dialysis, a blood bank, Level 3 imaging, and accredited residency training (DOH, 2012). A hospital can therefore be close by and still lack what a particular patient needs.

The third is bed reporting. While COVID-19 was spreading, the DOH used DataCollect to gather, from 1,906 public and private hospitals and infirmaries, the number of COVID-designated beds and mechanical ventilators each had in use. Reporting was 80 to 95 percent complete by April 2020 and 98 percent complete by December 2022 (Cabaro et al., 2023). Hospitals, in other words, will report bed counts through a regular channel. That channel, though, had firm backing from the national government and a purpose-built web platform behind it (Cabaro et al., 2023). Timely bed counts of this kind belong to what SDG target 3.d describes as a country's capacity for early warning and the management of health risks (UNGA, 2015).

1.2.8 Software Agents over E-mail

Wooldridge and Jennings (1995) describe an agent as software that runs on its own within an environment, notices and responds to changes there, acts on its own initiative to reach its goals, and communicates with other agents. Splitting a job among several agents means that one part can fail without stopping the rest. The study uses e-mail as its channel because every hospital already has an official address that its staff read. Messages go out through the Simple Mail Transfer Protocol (SMTP; Klensin, 2008), and replies are collected through the Internet Message Access Protocol (IMAP; Melnikov & Leiba, 2021), both of them open standards. A hospital has nothing to install, no account to create, and no system to connect to; it only has to reply.

1.2.9 Payment for Hospital Care in the Philippines

Three groups of laws largely decide what a patient pays and where. The first is the Universal Health Care Act, under which every Filipino is automatically enrolled in the National Health Insurance Program and basic or ward accommodation carries no co-payment (Republic Act No. 11223, 2019). The second is the Malasakit Centers Act. It places a Malasakit Center in every DOH hospital and in the Philippine General Hospital, a one-stop shop where poor and financially incapacitated patients can ask for medical and financial assistance, and it allows LGU hospitals and other public hospitals to open their own (Republic Act No. 11463, 2019). The third is Batas Pambansa Blg. 702 (1984), amended by Republic Act No. 8344 (1997) and Republic Act No. 10932 (2017). It penalizes any hospital or clinic that withholds basic emergency care in an emergency or serious case until a deposit or other advance payment is made. The first two laws tie the choice of hospital to cost, and they are national steps toward SDG target 3.8 on universal health coverage with protection from financial hardship (UNGA, 2015). The third takes cost out of that choice whenever the case is an emergency. Hospital mode is designed to respect all three.

1.3 Research Gap

The review points to three gaps on the transport side and a fourth that motivates the hospital extension.

- **(G1) No dataset.** No part of Albay has published, machine-readable routes, stops, or headways. Claims about accessibility in the province are therefore either absent or based on straight-line distance, which overstates access wherever terrain, water, or the road hierarchy forces a detour.
- **(G2) No measure of distribution.** Provincial and municipal plans describe transport mainly through road length and vehicle registrations. Neither figure tells a planner how many

important places a resident can reach in a given time, or how uneven that reach is from one place to another. Without such a measure the province cannot follow its progress on SDG target 11.2 or on the equity aims of SDG 10.

- **(G3) No model that knows the vehicle.** What a rural road in Albay can carry varies a great deal. Some roads are too narrow, too steep, or too rough, or they cross a bridge with a posted load limit; a provincial bus or a six-wheel truck cannot use them, although tricycles and cars travel on them every day. No open model records such limits, so freight and emergency trips depend on what individual drivers happen to know.
- **(G4) No live bed information and no routing by suitability.** Official registers record only licensed bed capacity, an administrative figure that says nothing about how many beds are free at a given moment. An ordinary map can show where hospitals are, but it cannot weigh travel time together with the level of care offered, current bed availability, and the patient's expected out-of-pocket cost. The hospital that is ready to admit a patient is often not the one nearest to the family or the ambulance crew. Closing this gap matters for SDG target 3.8, above all for low-income families who need prompt and affordable care.

Earlier work on informal transport relied on manual field mapping (Section 1.2.3). None has asked what can be proved mathematically when direct field collection is not possible, and that question is the focus of this study.

1.4 Contributions of the Study

- **C1.** Section 2 builds the mathematical framework that turns headway-based informal transit data into a time-expanded timetable graph, and it defines landmark-anchored travel-time accessibility and its spatial inequality around a 30-minute baseline budget.
- **C2.** Section 3.2 gives eight formal results with proofs. They establish the correctness and complexity of the query engine and a partial order on vehicle-restricted road networks, derive the structural properties of the inequality measures, and state the exact conditions under which regulated fares remain subadditive. Four of these results (Propositions 1, 2, 5, and 6) extend established theory to a landmark setting. The vehicle-ordering result, the fare-subadditivity conditions, and the corollaries that follow from them are the authors' own work.
- **C3.** Sections 3.3 to 3.5 run the computational pipeline on a complete end-to-end test instance, compare it with independently written reference implementations, and measure the errors caused by two common implementation shortcuts.
- **C4.** A repeatable procedure builds a landmark accessibility atlas and a location-checked GTFS feed from open sources only (Appendix A). Every input points back to a named public

register, so that others can audit or repeat the whole process, and the synthetic data can later be replaced by a field-verified feed without changing the mathematical model. The feed and the atlas also serve the open-data aims of SDG target 17.18.

The hospital extension adds two further contributions and four further results. Proposition 12 bounds the message load and the age of reports in the agent network; Proposition 10 shows that hospital suitability is monotone, and the payment-gap result (Corollary 11) follows directly from it; and Proposition 9 shows that stopping the hospital search early loses no exactness.

- **C5 (extension).** Section 2.8 describes a set of automated e-mail agents that ask registered hospitals for their current occupancy and free beds on a fixed schedule. The agents authenticate every reply and publish bed availability together with the age of each report. They were stress-tested over 30 simulated days in a mock mail environment.
- **C6 (extension).** Section 2.9 adds a hospital routing mode to the software tool. It matches the patient's acuity and ability to pay with each facility's service level and live bed count, and it sends patients and ambulances to the nearest suitable facility. Because Republic Act No. 10932 (2017) forbids holding back emergency care for payment, the payment filter is switched off in emergencies, so that, in line with SDG target 3.8, money never stands between a patient and emergency treatment.

1.5 The Study and the Sustainable Development Goals

The SDGs are interlinked, and progress on one goal can affect several others (UNGA, 2015). This study bears most directly on three of them. Because it is computational, it connects with these goals only as a measuring and planning tool: it has not been put into practice, and it reports no indicator value for Albay.

The closest link is with SDG 11, on sustainable cities and communities. Its target 11.2 calls for access to "safe, affordable, accessible and sustainable transport systems for all … with special attention to the needs of those in vulnerable situations" (UNGA, 2015). The landmark measure is related to indicator 11.2.1 (UNGA, 2017) but is not the same thing. The indicator is about reaching public transport itself, whereas the measure counts the essential services that public transport puts within 30 minutes. The fare model and the statutory discount address the affordability side of the same target.

The hospital extension answers to SDG 3, on good health and well-being. Hospital mode sends a patient to the nearest hospital that offers the required level of care, has a free bed, and fits what the patient can pay, and it never lets cost decide where an emergency patient goes. In this way it addresses target 3.8 on universal health coverage with financial risk protection. The bed-

availability network was built with target 3.d, on the management of health risks, in mind, and so far it has been tested only in simulation (Section 3.3).

SDG 10, on reduced inequalities, enters through the measures of Section 2.4. Targets 10.2 and 10.3 deal with inclusion and with more equal outcomes (UNGA, 2015). The Lorenz curve and the Gini, Theil, and Palma measures, together with the payment gap of Definition 19, show how unequal access is and where the inequality lies.

If the tool is validated and adopted, some of its features may also serve other goals: vehicle-class routing for infrastructure and disaster response (SDG 9 and target 11.5), the bed table during typhoons and Mayon Volcano alerts (SDG 13), the payment-gap map for planning services for the poor (SDG 1), the school layer (SDG 4), and the open dataset (SDG 17). These are possible uses, not demonstrated outcomes. Whether the study contributes to any of them in practice depends on field validation and on support from the LGUs and the provincial health office, both of which are left for future work (Section 4.3).

2. FRAMEWORK

2.1 Study Area, Landmarks, and Spatial Units

The study covers the whole province of Albay, that is, all 18 LGUs (Table 1) in its three legislative districts. The First District takes in the municipalities along the eastern seaboard; the Second District, the provincial capital, Legazpi City, together with the municipalities around it; and the Third District, the municipalities farther inland. Apart from Legazpi City, whose built-up area extends into neighboring Daraga, and the cities of Tabaco and Ligao, most LGUs are largely rural.

Urban, rural, coastal, inland, and island settings all occur within the province, so its residents face many different kinds of access problems in reaching school, work, trade, and health care. At the same time, the province is compact enough for a three-person team to study.

Table 1. The 18 LGUs of Albay, with 2020 population and land area. Population: PSA (2021). Land area: PhilAtlas (n.d.), compiled from PSA data.

District	Local Government Units	Population (2020)	Land Area (km²)
First	Tabaco City, Bacacay, Malilipot, Malinao, Santo Domingo, Tiwi	395,907	547.88
Second	Camalig, Daraga, Legazpi City, Manito, Rapu-Rapu	477,781	673.85
Third	Guinobatan, Jovellar, Libon, Ligao City, Oas, Pio Duran, Polangui	501,080	1,353.18
Total	**18 units**	**1,374,768**	**2,574.91**

The accessibility and inequality measures used below are standard, and each is cited where it first appears; what is new is the index set to which they are applied.

Definition 0 (Landmark set). A landmark is a fixed, named place of general public importance whose location can be pinned down and which appears in at least one publicly available register. The landmark set is

$$L = L_{\mathrm{a}} \cup L_{\mathrm{d}} \tag{1}$$

Source: formulated in this study (Definition 0)

where L_{a} is the set of anchors and L_{d} the set of essential destinations. Anchors are the places where people without a private vehicle are likely to begin a trip: a public transport terminal, the municipal or city hall, the public market, the town plaza, a port or pier, or a major junction where public transport operators usually pick up passengers. Essential destinations are places people go to for

the service they offer, such as hospitals and rural health units, public schools, government offices that deliver specific services, public markets, and designated evacuation centers. A place can be both an anchor and a destination; public markets usually are.

Each anchor $a \in L_a$ is represented by o_a, the road-network node closest to its register coordinate. Its demand weight P_a comes from the barangays assigned to it. Every barangay is first assigned to the anchor it can reach at the lowest cost over the road network, and the populations assigned to anchor a are then added together, using the counts of the 2020 census (PSA, 2021). P_a is therefore not an estimate. It is a sum of published barangay populations, all assigned by one and the same network rule.

Each destination $p \in L_d$ belongs to one service category $k \in K$ and carries a size w_p taken from the destination's own records, for example the service windows of an office, the classrooms of a school, or the reported beds of a hospital. $L^{(k)}$ denotes the destinations in category k. Hospitals need further attributes, because bed capacity alone does not tell whether a hospital can take a patient now; Section 2.7 lists them.

Table 2 names the government or official register behind each landmark category. Appendix A explains how the source data were obtained, and Appendix C gives the rules for admitting a record to the inventory and the fields kept for every landmark.

Table 2. Landmark categories, their role in the model, and the official registers used to identify them. Every category can be compiled from existing records without field visits.

Landmark Class	Role	Open Register
Public transport terminal	anchor	OpenStreetMap; route-plan annexes of LGUs
Municipal or city hall	anchor	OpenStreetMap; Philippine Standard Geographic Code
Town plaza, port, or pier	anchor	OpenStreetMap; port list of the Philippine Ports Authority
Major boarding junction	anchor	Highway junction nodes in OpenStreetMap
Hospital, infirmary, rural health unit	destination	National Health Facility Registry (DOH); hospital licensing records; PhilHealth accreditation list; Malasakit Center directory (DOH)
Public school (elementary and secondary)	destination	Masterlist of Schools, Department of Education
Government service office	destination	OpenStreetMap; agency directories
Public market	anchor and destination	OpenStreetMap; LGU lists
Designated evacuation center	destination	LGU disaster risk reduction plans

Sources: OpenStreetMap contributors (2026) and PSA (n.d.) for the anchors; for destinations, the Department of Education (2021), the DOH (n.d.-a, n.d.-b, 2012), and the Philippine Health Insurance Corporation (2026); ports from the Philippine Ports Authority (n.d.). Appendix A gives access details.

Barangays enter the model only through the populations that make up the anchor weights. No barangay is given an accessibility value of its own, no barangay centroid serves as an origin, and the paper makes no claim about any individual barangay.

2.2 Network Definitions

Definition 1 (Road network). $G_R = (V_R, E_R)$ is a weighted directed graph built from OpenStreetMap (OpenStreetMap contributors, 2026). Each arc e has a length $\ell_e > 0$ and a free-flow speed v_e, and its attribute vector a_e gives four limits for the arc: maximum axle load, maximum gross mass, vertical clearance, and usable width. An attribute that was never surveyed is set to $+\infty$ and flagged.

Definition 2 (Walking network). $G_W = (V_R, E_W)$ uses the same vertices as the road network. Walking an arc takes ℓ_e / v_{walk} with $v_{walk} = 4.5$ km/h, a design value that falls within the normal adult walking speeds reported by Bohannon and Williams Andrews (2011). A single walking leg is limited to $\bar{t}_W = 15$ min.

Definition 3 (Transit network). Boarding points form the set S. One run of a vehicle visits a sequence of stops, and that sequence is a trip. A connection $c = (c_{dep}, c_{arr}, \tau_{dep}(c), \tau_{arr}(c), \text{trip}(c))$ records that a vehicle leaves stop c_{dep} at time $\tau_{dep}(c)$ and reaches the next stop c_{arr} at time $\tau_{arr}(c)$. All connections of the service day form the set C. Changing to a vehicle on a different trip requires a minimum change time θ_{ch}.

Definition 4 (Time-expanded graph, following Pyrga et al., 2008). The time-expanded graph is $G = (V, E)$, in which each vertex is a stop–time pair (s, t) standing for a departure, arrival, or transfer event. There are four kinds of arcs: ride arcs, from the departure event of a connection to the arrival event of the same connection; wait arcs, between consecutive events at the same stop; transfer arcs of duration θ_{ch}, from an arrival event to the earliest compatible departure event at that stop; and walk arcs, from an arrival event at s to a departure event at s', with duration equal to the walking time. Every arc satisfies $t(\text{head}) \geq t(\text{tail})$.

2.3 Travel Time and Accessibility

Definition 5 (Earliest arrival, following Pyrga et al., 2008). Fix an origin o, a destination d, and a departure time t_0, and let $J(o, d, t_0)$ be the set of feasible journeys from o to d that leave no earlier than t_0. Then

$$\tau^*(o, d, t_0) = \min \{ \text{ arrival time of } j : j \in J(o, d, t_0) \} \tag{2}$$

Source: Pyrga et al. (2008)

with $\tau^* = +\infty$ when $J(o, d, t_0)$ is empty. The travel time from door to door is $T(o, d, t_0) = \tau^*(o, d, t_0) - t_0$. Because informal operators publish no departure times, T depends on which departure instant happens to be sampled, and the atlas therefore reports averages over a grid of departure times. Owen and Levinson (2015) did the same when they averaged transit accessibility over every minute of a departure window, and Stępniak et al. (2019) showed that accessibility estimates shift with the temporal resolution used. Averaging deals with what is, in effect, a clock-time version of Openshaw's (1984) modifiable areal unit problem:

$$\bar{T}(o, d) = \frac{1}{|\mathcal{T}_0|} \sum_{t_0 \in \mathcal{T}_0} T(o, d, t_0), \qquad \mathcal{T}_0 = \{06{:}00, 06{:}05, \ldots, 08{:}00\} \tag{3}$$

Source: averaging, as in Owen and Levinson (2015); see also Stępniak et al. (2019)

Definition 6 (Cumulative landmark accessibility, adapted from Wachs & Kumagai, 1973). Let k be a destination type of Table 2. For an anchor a, a type k, and a budget θ,

$$A_a^k(\theta) = \sum_{p \in L^{(k)}} w_p \cdot \mathbf{1}[\, \bar{T}(o_a, p) \le \theta \,] \tag{4}$$

Source: adapted from Wachs and Kumagai (1973); the anchor replaces the residential zone.

where w_p is the size of destination p in service windows, classrooms, or beds. Equation (4) keeps the form of the cumulative opportunity measure of Wachs and Kumagai (1973). Only the origin is different, since an anchor of Definition 0 takes the place of the residential zone. The main baseline is $\theta = 30$ min, and Section 3.6 tries other budgets in its sensitivity analysis. Read in this way, the measure complements SDG indicator 11.2.1, the "proportion of population that has convenient access to public transport" (UNGA, 2017): the indicator counts who can reach public transport, while $A_a^k(\theta)$ counts what that transport actually reaches.

Definition 7 (Impedance-weighted accessibility, based on W. G. Hansen, 1959). As a robustness check, a second form discounts each destination by a negative exponential of its travel time:

$$A_a^{k,\beta} = \sum_p w_p \exp(-\beta \bar{T}(o_a, p)), \qquad \beta > 0 \tag{5}$$

Source: W. G. Hansen (1959); exponential form as reviewed by Handy and Niemeier (1997)

Equation (5) is W. G. Hansen's (1959) gravity measure with exponential impedance, one of the standard functional forms reviewed by Handy and Niemeier (1997). Since it has no threshold, Equation (5) is reported alongside the cumulative measure wherever a result is affected by the sharp step discussed in Section 3.5.

2.4 Measures of Access Inequality

Consider a single destination type. Let $A = (A_1, \ldots, A_{Na})$ be the vector of anchor accessibility scores with anchor weights P, and write $W = \Sigma_a P_a$ for the total weight and $\mu = \Sigma_a P_a A_a / W$ for the weighted mean.

Definition 8 (Population-weighted Gini coefficient, from Gini, 1912). The weighted form below is the one given by Cowell (2011) and used by Pereira et al. (2017) and Lucas et al. (2016) to measure inequality.

$$G(A, P) = \frac{\sum_i \sum_j P_i P_j |A_i - A_j|}{2W^2 \mu} \tag{6}$$

Source: Gini (1912), in the weighted form of Cowell (2011)

Definition 9 (Theil index, from Theil, 1967). Let $p_a = P_a / W$. Then

$$T(A, P) = \sum_a p_a \frac{A_a}{\mu} \ln \frac{A_a}{\mu}, \qquad A_a \geq 0, \;\; 0 \ln 0 = 0 \tag{7}$$

Source: Theil (1967)

For the Lorenz (1905) curve of access, anchors are sorted by A_a, and the cumulative share of weight is plotted against the cumulative share of reachable destinations. The Palma ratio (Cobham & Sumner, 2013; Palma, 2011) is reported as well. It divides the access of the best-served tenth of the weighted population by that of the worst-served four-tenths, and readers without statistical training usually find it easier to interpret than a Gini value (Cobham & Sumner, 2013). Through these four measures the study reads SDG 10 in terms of access. Targets 10.2 and 10.3 concern inclusion

regardless of age, disability, or economic status and the reduction of unequal outcomes (UNGA, 2015), and a Gini, Theil, or Palma value computed over anchors shows how far access to essential places still is from that aim.

2.5 Vehicle Classes on the Road Network

Definition 10 (Vehicle class and feasible subgraph). A vehicle class is a vector $\gamma = (\gamma^{w}, \gamma^{h}, \gamma^{m}, \gamma^{x})$ whose components are, in order, the width the vehicle needs, the clearance it needs, its gross mass, and its load per axle. The vehicle may use the arcs in

$$E_{\gamma} = \{\, e \in E_{R} : a^{w}_{e} \geq \gamma^{w},\ a^{h}_{e} \geq \gamma^{h},\ a^{m}_{e} \geq \gamma^{m},\ a^{x}_{e} \geq \gamma^{x} \,\} \tag{8}$$

Source: formulated in this study (Definition 10)

so that $G_{\gamma} = (V_{R}, E_{\gamma})$ is its feasible subgraph, with shortest-path distance d_{γ}. Writing $\gamma \leq \gamma_{0}$ means that the inequality holds in every component. Definition 10 and Proposition 3 were formulated by the researchers.

Every component of the sedan vector is bounded above by the matching component of the sport utility vehicle (SUV). Sedans and motorized tricycles, by contrast, cannot be compared under this order: a sedan needs more road width, while a tricycle, because of its canopy, needs more overhead clearance. Section 3.2 works out what these limits mean for routing, and Remark 2 gives the order in which the road graphs should be preprocessed. For hospital mode, a further class γ_{amb} represents a van-type ambulance, with entries set as design values. Because routes respect each vehicle's physical limits, relief trucks and ambulances can be kept on roads they can actually use after a strong typhoon or an eruption of Mayon Volcano. This supports target 9.1 on infrastructure that serves everyone and target 11.5 on reducing disaster losses (UNGA, 2015).

Table 3 lists the vehicle classes used for routing. None of the entries is a legal limit; all are representative design values chosen by the researchers. Pedestrians are given a nominal point load of 0.1 t, and axle load equals gross mass divided by the number of axles.

Table 3. Vehicle classes used for routing (design values set by the researchers, not legal limits).

Class	Width (m)	Clearance (m)	Gross Mass (t)	Axles	Axle Load (t)
Pedestrian	0.8	2.0	–	–	–
Motorcycle, tricycle	1.3	2.0	0.6	2	0.30
Sedan	1.8	1.6	2.0	2	1.00
Sport utility, pickup	2.0	2.0	3.0	2	1.50

Class	Width (m)	Clearance (m)	Gross Mass (t)	Axles	Axle Load (t)
Ambulance (van type)	2.0	2.6	3.5	2	1.75
Jeepney, UV van	2.1	2.6	4.5	2	2.25
Four-wheel truck	2.2	3.2	7.5	2	3.75
Six-wheel truck	2.5	3.8	15.0	3	5.00
Ten-wheel truck	2.5	4.2	26.0	5	5.20
Provincial bus	2.6	4.0	18.0	3	6.00

2.6 Fare Model

Definition 11 (Regulated distance fare; Daguno-Bersamina, 2026; Nieto, 2026). On a mode whose base fare f_0 covers the first d_0 kilometers and whose rate for each succeeding kilometer is r, a leg of d kilometers costs

$$f(d) = f_0 + r \cdot \lceil \max(0,\, d - d_0) \rceil \tag{9}$$

Source: fare structure and parameters reported by Daguno-Bersamina (2026) and Nieto (2026); see Table 4.

Any fraction of a succeeding kilometer is charged as a whole kilometer, a rounding convention chosen for this study. Senior citizens, students, and persons with disabilities receive the statutory 20 percent discount reviewed in Section 1.2.6, so their fare is $f_{\delta}(d) = (1 - \delta) f(d)$ with $\delta = 0.20$. A journey J whose m legs have lengths $d_1, \ldots, d_m$ costs

$$F(J) = \sum_{i=1}^{m} f(d_i) \tag{10}$$

Source: formulated in this study

Table 4. Fare parameters in effect from September 28, 2026, as approved by the LTFRB and reported by Daguno-Bersamina (2026) and Nieto (2026). Tricycle fares follow local ordinances and are collected by municipality.

Mode	Base Fare f_0 (₱)	Base Distance d_0 (km)	Rate r (₱/km)
Traditional jeepney	14.00	4	2.00
Modern jeepney	17.00	4	2.40
Provincial ordinary bus	12.00	5	2.20
Tricycle	set by ordinance	set by ordinance	set by ordinance

2.7 Extension: Hospital Records, Capability, and Cost Class

Sections 2.7 to 2.9 extend the model to hospital routing. Nothing in Sections 2.1 to 2.6 changes: the extension reads the same graph, the same vehicle classes, and the same landmark set. It works on a subset of the essential destinations of Definition 0 and needs to know more about each of them than its size.

Definition 12 (Hospital record). Let $H \subseteq L_d$ be the hospitals and infirmaries among the destinations. Each $h \in H$ carries its entrance node v_h (Definition 0); its authorized bed capacity c_h; a service capability level $\eta_h \in \{0, 1, 2, 3\}$, where 0 denotes an infirmary and 1 to 3 the hospital levels of Table 5; an emergency-room flag $e_h \in \{0, 1\}$; a cost class $q_h \in \{0, 1, 2, 3\}$; and a contact address m_h, the e-mail address that the facility publishes for official correspondence. The cost class is assigned by the rule

$$q_h = \begin{cases} 0 & \text{if } h \text{ is government-owned and hosts a Malasakit Center,} \\ 1 & \text{if } h \text{ is government-owned without a Malasakit Center,} \\ 2 & \text{if } h \text{ is privately owned and PhilHealth-accredited,} \\ 3 & \text{if } h \text{ is privately owned and not PhilHealth-accredited.} \end{cases} \tag{11}$$

Source: this study; program flags from Republic Acts No. 11223 and 11463 (2019)

The ordering ranks hospitals by expected out-of-pocket burden; it is not a price. Its inputs (ownership, PhilHealth accreditation, and the presence of a Malasakit Center) are all published (Appendix A), so any reader can check the class of any hospital. Table 6 pairs each cost class with the payment class that a patient selects in the application (Definition 15).

Table 5. Service capability levels in the model, following the DOH (2012) classification of hospitals.

η_h	Facility	Services Relevant to Routing
0	Infirmary	Placed below Level 1 in the model and offered only to profiles whose minimum level is 0
1	Level 1 hospital	Level 1 imaging, pharmacy and blood station, secondary clinical laboratory, isolation and maternity facilities, and an operating room with a recovery room
2	Level 2 hospital	Everything at Level 1, plus a general ICU and a neonatal ICU, care for high-risk pregnancy, a tertiary clinical laboratory, and Level 2 imaging
3	Level 3 hospital	Everything at Level 2, plus dialysis, a blood bank, physical medicine and rehabilitation, accredited residency training, and Level 3 imaging, including interventional radiology

Table 6. Cost classes (Equation 11) and the payment classes users choose from. Payment class ρ admits every hospital with $q_h \le \rho$.

Class	Hospital in Cost Class q_h	Patient's Choice in Payment Class ρ	Hospitals Accepted
0	Public hospital that hosts a Malasakit Center	"Free or assisted care only"	$q_h = 0$
1	Public hospital with no Malasakit Center	"Public hospitals"	$q_h \le 1$
2	Private, PhilHealth-accredited hospital	"Private hospitals that accept PhilHealth"	$q_h \le 2$
3	Private hospital without accreditation	"Any hospital"	$q_h \le 3$

2.8 Extension: Agent Network for Bed Availability

The registers of Appendix A record authorized beds, not free ones. To learn how many beds are free, the researchers built six kinds of software agents that write to the hospitals and ask (Table 7, Figure 1). Here an agent is a small program with a single job. All six read from one database, which only the Registry agent may change, and they pass work to one another through a single message queue. One job is left to a person because no fixed rule can do it safely: deciding what to do with a reply whose meaning the rules cannot settle. The live table of free beds that the agents maintain is the part of the application aimed at SDG target 3.d.

Table 7. Roles in the bed-availability network.

Agent	Task
Registry	Holds the hospital records of Definition 12, the opt-out list, and the bed-status table. No other agent may write to the database.
Coordinator	Opens each inquiry cycle on the schedule of Definition 13, enforces the opt-out list and the message limits, hands inquiries to the inquiry agents, and decides when a reminder is due.
Inquiry (one per legislative district)	Prepares the inquiry of Appendix F, attaches a one-time reference token, and sends it by SMTP to the hospital's contact address.
Reader	Checks the project mailbox over IMAP every 10 minutes and once more just before any send. It records opt-outs, matches each reply to its token and sender, and extracts the figures by fixed rules. It never guesses: anything it cannot read with certainty goes to the human review queue.
Validator	Tests each reply against rules V1 to V5 of Table 8 and passes accepted reports to the Registry.

Agent	Task
Status	On request from the application, computes bed status, free beds, and report age (Definition 14).
Human reviewer (a person, not an agent)	A researcher who clears the review queue. A figure entered by hand passes through the Validator exactly like any other figure.

Definition 13 (Inquiry protocol). For hospital h, cycle k opens at

$$t_k = t_1 + (k-1)P_h, \text{ with reminders only at } t_k + jR_{rem},\ 1 \le j \le n_{rem},\ n_{rem}R_{rem} < P_h \qquad (12)$$

Source: this study (Definition 13)

By default $P_h = P_{cyc} = 8$ h, which gives one cycle per standard nursing shift; a facility that prefers a single daily inquiry is given $P_h = 24$ h. When a cycle opens, the Coordinator sends an inquiry with a fresh token to every hospital not on the opt-out list. If no validated reply quoting that token has arrived by $t_k + R_{rem}$, one reminder ($n_{rem} = 1$) goes out with the same token, and nothing more is sent until the next cycle. A hospital may send an update at any time by quoting any active token, and tokens stay valid for 24 h so that late replies still count. Opting out is automatic: once a reply contains a recognized opt-out phrase, the hospital is left out of every later cycle. Just before any send, the Reader processes every message waiting in the mailbox.

Figure 1. Structure of the bed-availability network. Solid arrows show inquiries, replies, and reports; dashed arrows show opt-outs and figures entered after human review.

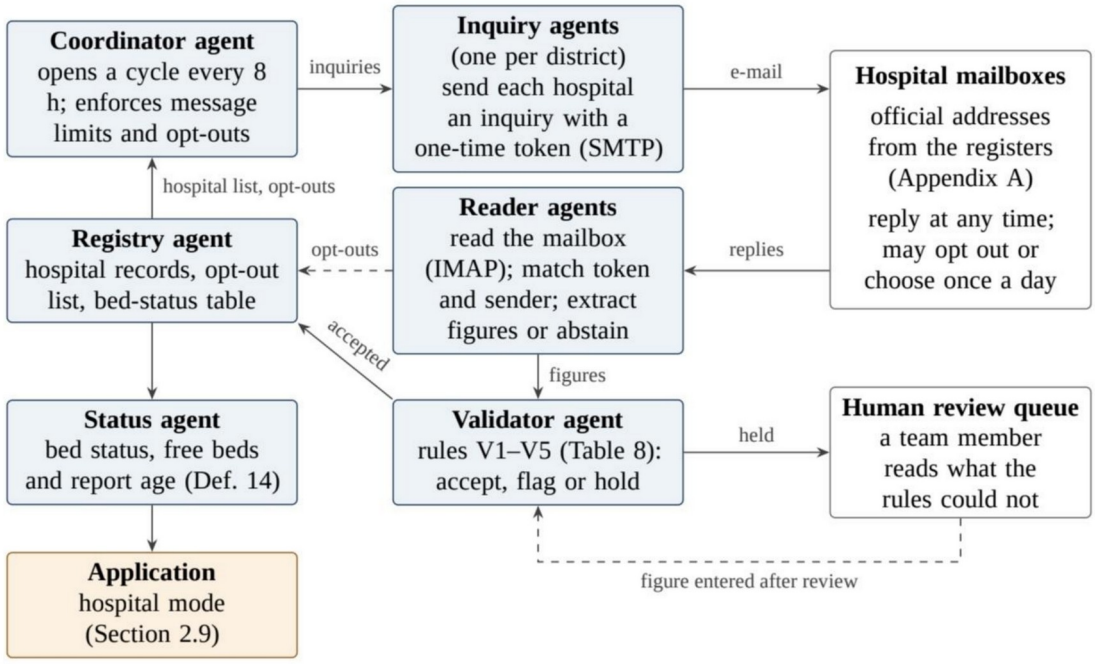


Table 8. The five checks that the Validator agent runs on every reply.

Rule	Name	Check	If the Check Fails
V1	Authenticity	The subject line carries a token issued to *h* less than 24 h earlier, and the sender is the registered address m_h.	Discarded; nothing stored
V2	Completeness	The reply yields one capacity figure and one figure for either occupied or free beds (occupied beds are derived in the second case). Contradictory figures count as unreadable.	Referred to human review
V3	Capacity plausibility	$0.5c_h \le \hat{c}_h \le 2c_h$, where $\hat{c}_h$ is the capacity stated in the reply.	Held, as a possible typing error or a different notion of a bed
V4	Occupancy plausibility	$0 \le \hat{o}_h \le 2\hat{c}_h$. A count above $\hat{c}_h$ is treated as "over capacity."	Held
V5	Recency	The reply is newer than the report already on file.	Added to the history only

Rule V4 lets reported occupancy exceed nominal capacity because, during a surge, patients are often placed in shared beds or on hallway stretchers. Such a count reflects real operating conditions, so the hospital is treated as full instead of the reply being rejected.

Definition 14 (Bed status and report freshness). For hospital *h*, a validated report is a tuple $r_h = (t^r_h, \hat{c}_h, \hat{o}_h, \hat{e}_h)$: the time of the reply, the reported total bed capacity, the reported number of occupied beds, and, optionally, the reported emergency-room status. From the latest validated report,

$$b_h(t) = \max(0,\ \hat{c}_h - \hat{o}_h), \qquad age_h(t) = t - t^r_h \tag{13}$$

Source: this study (Definition 14)

$$\sigma_h(t) = \begin{cases} \text{UNKNOWN} & \text{if no validated report exists or } age_h(t) > age_{\max}, \\ \text{FULL} & \text{if } age_h(t) \le age_{\max} \text{ and } (b_h(t) = 0 \text{ or } \hat{e}_h = \text{FULL}), \\ \text{AVAILABLE} & \text{otherwise,} \end{cases} \tag{14}$$

Source: this study (Definition 14)

where $age_{\max}$ = 12 h. The application never shows a status without the age of its report, as in "9 free beds, reported 1 h 40 min ago."

2.9 Extension: Finding the Nearest Suitable Hospital

In hospital mode the application becomes a routing tool for ambulance crews and for patients who travel on their own. The map keeps only the hospitals and infirmaries of Definition 12 and marks

those that fit the patient's clinical and payment profile, so the user can see how far away each suitable facility is. Figure 2 summarizes the whole workflow.

Definition 15 (Patient profile). A profile $\pi = (\kappa, \rho)$ combines an acuity level $\kappa \in \{1, \ldots, 5\}$ from the Emergency Severity Index, a five-level triage scale on which level 1 is the most urgent (Gilboy et al., 2011), with a payment class $\rho \in \{0, 1, 2, 3\}$ from Table 6, where $\rho_{\max} = 3$. The ambulance crew enters the acuity level, or the user picks it from the plain descriptions in Table 9; the application itself never triages. Each acuity level κ is paired with a minimum service level $\lambda(\kappa)$. Levels 1 and 2 form the emergency set $K_E = \{1, 2\}$, which stands for what Republic Act No. 10932 (2017) calls "emergency or serious cases." A trip uses one of two modes: x = amb, an ambulance on the road network, or x = pt, a patient riding public transport and walking.

Table 9. Minimum service level and emergency flag for each acuity level. Descriptions adapted from Gilboy et al. (2011); λ and K_E are declared design parameters.

κ	Description	$\lambda(\kappa)$	In K_E
1	Life-saving treatment needed at once	3	yes
2	High-risk situation; new confusion, lethargy, or disorientation; or severe pain or distress	2	yes
3	Stable; probably needs two or more hospital resources	1	no
4	Stable; probably needs one resource	1	no
5	Stable; probably needs no resources	0	no

Table 9 is a placeholder until the provincial health office issues a formal referral policy. Section 3.2 relies on only one property of λ: it is non-increasing in the acuity level κ, so a more urgent case never lowers the required service level.

Definition 16 (Suitable hospital). At time t, hospital h suits profile π when $\Phi(h; \pi, t) = 1$, where

$$\Phi(h; \pi, t) = 1[\eta_h \geq \lambda(\kappa)] \cdot 1[\kappa \notin K_E \text{ or } e_h = 1] \cdot 1[\sigma_h(t) \neq \text{FULL}] \cdot 1[\kappa \in K_E \text{ or } q_h \leq \rho] \quad (15)$$

Source: formulated in this study; service levels from DOH (2012), acuity levels from Gilboy et al. (2011), emergency exception derived from Republic Act No. 10932 (2017).

The set of suitable hospitals is $H_\pi(t) = \{\, h \in H : \Phi(h; \pi, t) = 1 \,\}$.

The four factors of Equation (15) test, in turn, capability, emergency readiness, available room, and affordability. Two design choices need explaining. First, a hospital whose status is UNKNOWN stays in $H_\pi(t)$. An unanswered message or an administrative delay does not mean the beds are full, and dropping such hospitals would penalize nearby patients for a clerical lapse; the application therefore lists UNKNOWN hospitals together with the age of their last report. Second, the payment

factor equals 1 for emergency profiles ($\kappa \in K_E$), whatever the patient can pay. This follows Philippine law, since Republic Act No. 10932 (2017) bars hospitals from demanding a deposit or other advance payment before giving basic emergency care. Under that law, whether a case is an emergency is for the attending physician to decide; the acuity level in the model only guides routing and does not stand in for clinical judgment. By confining the payment filter to non-emergency trips, Equation (15) builds SDG target 3.8 into the routing rule.

Definition 17 (Nearest suitable hospital). For an origin o, a time t, and a profile π, let $T_x(o, h, t)$ be the travel time from o to hospital h when leaving at t in mode x. In ambulance mode (x = amb), it is the shortest-path time from o to the entrance node v_h over the ambulance subgraph $G_{\gamma amb}$ of Definition 10, with arc times ℓ_e/v_e. In public transport mode (x = pt), a sink vertex z_h is added for each hospital to the graph of Definition 4. The sink is reached from every arrival event at a stop within walking distance of h, by an arc whose duration equals the walk to the hospital, and $T_{pt}(o, h, t) = \tau^*(o, z_h, t) - t$. For profile π, the nearest suitable hospital and the shortest travel time are

$$h^*(o,\pi,t) = \arg\min_{h \in H_\pi(t)} T_x(o,h,t), \qquad T^*(o,\pi,t) = \min_{h \in H_\pi(t)} T_x(o,h,t) \tag{16}$$

Source: formulated in this study (Definition 17)

Ties go to the hospital with more reported free beds $b_h(t)$ and, if a tie remains, to the smaller identifier. If no suitable hospital can be reached within the network, $T^* = +\infty$. In the application, the origin o is the user's GPS position snapped to the nearest road segment or stop, and t is the current time. A single query thus uses one departure time; the averaging grid of Equation (3) is used only for atlas summaries.

Definition 18 (Stabilization rule, a design choice). The rule applies only when $\kappa = 1$. Ignoring service level and cost class, let $h°$ be the nearest hospital with an open emergency room ($e_h = 1$) that is not FULL. If

$$T_x(o, h°, t) + \theta_{stab} < T^*(o, \pi, t), \qquad \theta_{stab} = 15 \text{ min} \tag{17}$$

Source: formulated in this study (Definition 18)

the application returns $h°$ first, with the message "stabilize, then transfer to h^*," and h^* second. A patient who needs immediate life-saving treatment should not be driven past an open emergency room on the way to a Level 3 hospital. The margin θ_{stab} stands in for the provincial referral protocol and, like Table 9, is a placeholder until the provincial health office sets a formal policy.

Definition 19 (Payment gap and bed accessibility). At origin o, the payment gap of profile (κ, ρ) is

$$\Delta(o, \kappa, \rho, t) = T^*(o, (\kappa, \rho), t) - T^*(o, (\kappa, \rho_{\max}), t), \quad \text{with } \infty - \infty := 0 \tag{18}$$

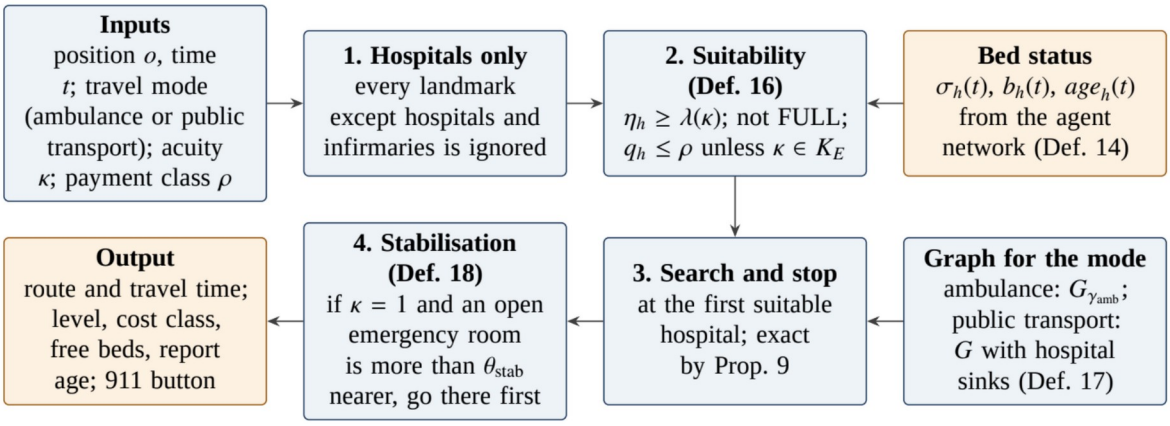
Source: this study (Definition 19)

that is, the extra travel time imposed by a limited ability to pay. The bed counterparts of Definition 6 count, for each anchor, the reported free beds and the free beds in hospitals that payment class ρ accepts:

$$B_a(\theta, t) = \sum_{h \in H} b_h(t)\, \mathbf{1}[\bar{T}(o_a, h) \le \theta], \qquad B_a^{\rho}(\theta, t) = \sum_{h \in H:\, q_h \le \rho} b_h(t)\, \mathbf{1}[\bar{T}(o_a, h) \le \theta] \tag{19}$$

Source: formulated in this study by applying Wachs and Kumagai (1973) to free beds

Figure 2. Steps of hospital mode, from inputs to recommendation. Orange boxes depend on the agent network.



2.10 Assumptions

- **Assumption 1 (Nonnegative traversal).** No arc of G has t(head) < t(tail), so a journey never goes back in time.
- **Assumption 2 (No overtaking within a route).** On any route, a vehicle that departs earlier never arrives later than one that departs after it.

Together, these give the first-in, first-out (FIFO) property that shortest-path search on time-dependent networks needs in order to remain exact (Kaufman & Smith, 1993; Pyrga et al., 2008). Six further assumptions complete the model.

- **Assumption 3 (Service frequency).** Routes without a timetable are given departures spaced at the average headway. The sensitivity runs compare the waiting times this produces with a half-headway waiting model.

- **Assumption 4 (No waiting at the destination).** A traveler who reaches a landmark is served there without queuing or capacity limits. This is standard practice for cumulative accessibility (Geurs & van Wee, 2004), and its consequences are discussed in Section 4.2. Hospital mode relaxes the assumption by reading live bed counts, but it still does not model competition for resources, such as two ambulances heading for the same last bed (Definition 14).
- **Assumption 5 (Open data only).** Every number in this paper comes from one of four traceable sources: a mathematical derivation, a fixed model parameter, a public administrative record (Appendix A), or a synthetic data set made only for software testing. No interview with any facility and no physical field measurement was carried out, and no number comes from one.
- **Assumption 6 (Reported capacity).** In live operation, bed figures come from the hospitals themselves. Replies are checked for format and range (Table 8), but the reported capacity is not verified in any other way at this stage.
- **Assumption 7 (Emergency care).** Hospitals with an emergency room are assumed to follow Republic Act No. 10932 (2017), which requires them to give basic emergency care in emergency or serious cases without first asking for a deposit or advance payment. This is why the payment filter is dropped for $\kappa \in K_{\mathrm{E}}$.
- **Assumption 8 (Ambulance travel time).** Ambulance travel times use the free-flow speeds of Definition 1. Dispatch time, handover time, emergency driving, and traffic delays are not modeled.

2.11 Notation

Table 10. Symbols and their meanings.

Symbol	Meaning
L, L_{a}, L_{d}	all landmarks; anchors; essential destinations
N_{a}	number of anchors
P_{a}, W	weight of anchor a; sum of all weights
S, C	stops; connections
$G = (V, E)$	the time-expanded timetable graph
θ_{ch}	minimum change time between trips
$\tau^{*}(o, d, t_0)$	earliest arrival
T, $\bar{T}$	travel time from door to door; its mean over departure times
θ	travel-time budget

Symbol	Meaning
$A_a^k(\theta)$	total size of type-k destinations reachable from anchor a within θ
μ	weighted mean accessibility
G, T	Gini coefficient; Theil index
γ, E_γ, d_γ; γ_{amb}	vehicle class, arcs open to it, its shortest distance; class of the ambulance
$f(d)$, $F(J)$	fare for a leg; fare for a journey
H, v_h, c_h	hospitals; entrance node; authorized beds
η_h, e_h, q_h, m_h	service level; emergency-room flag; cost class; contact address
P_h, R_{rem}, n_{rem}, D_{rep}	inquiry period; reminder delay; reminders allowed per cycle; longest reply delay assumed
$\hat{c}_h$, $\hat{o}_h$, $\hat{e}_h$, $b_h(t)$	capacity, occupied beds, and emergency-room status as reported; free beds
$\sigma_h(t)$, $age_h(t)$, age_{max}	bed status; age of the report; limit on age
$\pi = (\kappa, \rho)$	patient profile made of acuity level and payment class
$\lambda(\kappa)$, K_E	lowest acceptable service level; emergency set
Φ, $H_\pi(t)$	indicator of suitability; set of suitable hospitals
T_x, h^*, T^*	travel time in mode x; nearest suitable hospital; time to reach it
$h°$, θ_{stab}	nearest open emergency room; margin for stabilization
Δ, B_a, B^ρ_a	payment gap; accessibility to free beds; free beds that payment class ρ accepts

Sources of the equations. Equations (2) to (7), (9), (20), and (21) are established constructions, each cited where it first appears. The rest, Equations (1), (8), (10) to (19), (22), and (23), together with Definitions 0, 10, and 12 to 19, were formulated by the researchers; Equation (19) applies Wachs and Kumagai (1973) to free beds. In Section 3.2, the headings of Propositions 1, 2, 5, and 6 name the sources of those known results, whose proofs the researchers rewrote for landmarks, while Corollaries 4, 8, and 11 and Propositions 3, 7, 9, 10, and 12 are new. Service levels (Definition 12) follow the DOH (2012) classification and acuity levels (Definition 15) follow Gilboy et al. (2011), but the way they are combined with bed status and cost is the study's own. Wherever a known measure was adapted, only the index set changed (anchors in place of residential zones or income earners, and free beds in place of authorized beds); the algebra was left untouched.

3. FINDINGS

3.0 Note on Data Sources and Integrity

Every result in this section was produced with simulated data alone, and no hospital was contacted. Each number belongs to one of three kinds, and the text says which: a figure taken from a published source listed in Appendix A; a value fixed by a rule stated in Section 2, such as the walking speed, the minimum change time, or the acuity table; or an output computed on a synthetic instance built for testing. All results in Sections 3.2 to 3.5 were checked against independent solutions of synthetic instances. The mail agents of Section 2.8 were run against a mock mail server, with mock messages sent from the test machine. The hospitals in that test are not real, and no real bed count is involved. All simulated routes, timetables, accessibility values, bed states, and inequality values are model outputs and are labeled as such wherever they appear. They say nothing about transport or health facilities in Albay and should not be quoted as if they did. The same holds for Section 3.5: instance S1 is a simulated seven-stop test network, and the hospital attributes in Table 20 were invented for demonstration and do not describe Albay.

3.1 General Methods

The work proceeded in eight stages, none of which involved a manual survey or a field visit. The hospital extension occupies Stages 6 and 7.

Stage 1: Road network. The network of Definition 1 was built from OpenStreetMap road data (OpenStreetMap contributors, 2026). On this geometry the model generates candidate routes that connect the main market centers and transit hubs of Table 2 while respecting the physical limits of each vehicle class.

Stage 2: Landmark screening and mapping. Candidate landmarks were drawn from the registers of Table 2 and screened with the rules of Appendix C. After duplicates were removed, each landmark was snapped to its nearest road access node. Generated routes and candidate stops were then compared with the available spatial data against validation thresholds fixed in advance. A landmark whose published coordinate did not fit the road network was flagged and dropped, never moved by hand.

Stage 3: GTFS feed generation. A standard GTFS feed (Appendix B) was written with stops, routes, trips, stop times, calendar rules, fare rules, and route shapes. Before work continued, the feed had to pass the open-source GTFS Schedule validator (MobilityData, n.d.-b) with no blocking errors.

Stage 4: Graph construction and queries. The feed and the walking links were combined into the time-expanded graph of Definition 4, and earliest arrival times (Definition 5) were computed for every origin–destination pair at every departure time of the grid.

Stage 5: Accessibility and inequality. Cumulative landmark accessibility was computed from the shortest travel times, and the resulting distributions were summarized with Lorenz curves, Gini coefficients, Theil indices, and Palma ratios.

Stage 6: Hospital layer. Every hospital received a record as in Definition 12. Ownership, service level, and emergency-room status came from DOH records, and contact addresses from official published hospital records. PhilHealth accreditation and Malasakit Center flags were taken from the Appendix A sources, and the cost class was then set by Equation (11).

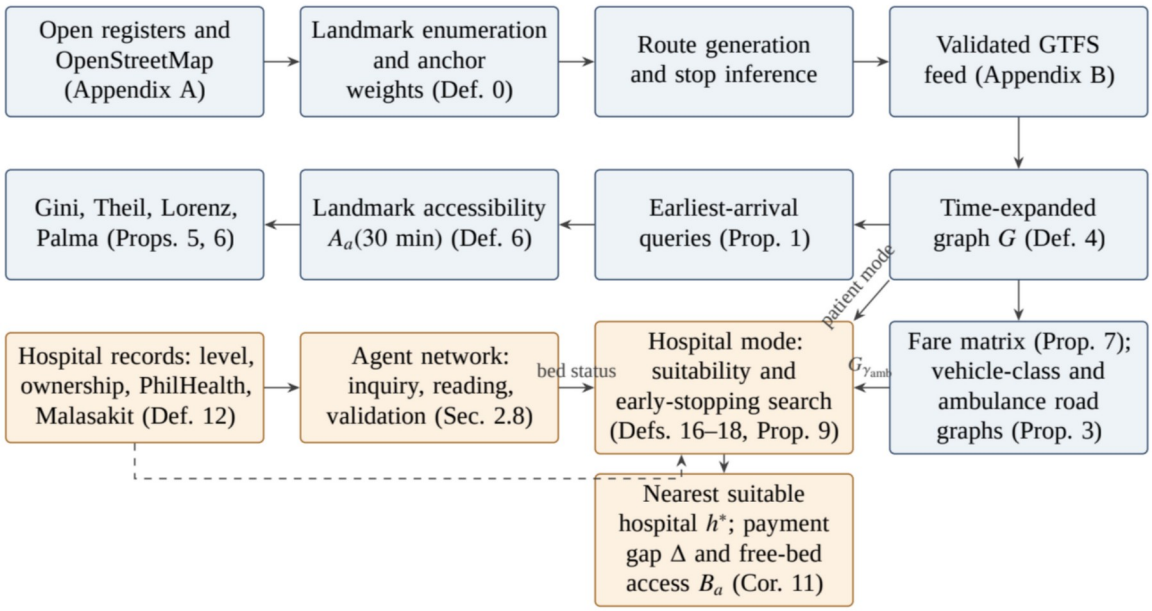
Stage 7: Live bed tracking. Following the protocol of Definition 13, the e-mail agents of Section 2.8 publish, for each facility, the reported free beds, the bed status, and the age of the report (Definition 14). This stage was exercised only in a simulated mail environment (Section 3.3).

Stage 8: Prototype. The routing engine, the accessibility outputs, and the GTFS feed were combined into a prototype web application. It offers route directions, stop locations, fare calculation, vehicle-restricted routing, and a separate hospital mode (Section 2.9; Appendix D).

3.2 Theoretical Results

The correctness of the query engine has to be settled before any accessibility analysis can be trusted. Figure 3 shows where each theoretical result sits in the processing pipeline.

Figure 3. Place of each theoretical result in the processing pipeline.



The query stage takes roads from OpenStreetMap, facility records from government registers, and census counts from the PSA. Fares and vehicle-class routes are computed on the graph, and hospital mode combines the graph with hospital records and agent-reported bed status.

Proposition 1 (Query engine correctness; after Dijkstra, 1959; Pyrga et al., 2008). *Let Assumption 1 hold. For any origin o, destination d, and departure time t_0, a label-setting shortest-path search on the time-expanded graph G that uses absolute arrival time as its label and settles vertices in nondecreasing order of label returns $\tau^*(o, d, t_0)$ exactly.*

Proof. Give each arc (u, v) of E the weight $\omega(u, v) = t(v) - t(u)$; by Assumption 1, every such weight is nonnegative. Along any path π from the source (o, t_0) to a vertex (d, t), the weights telescope, $\Sigma_{(u,v)\in\pi}\, \omega(u, v) = t - t_0$, so a path has minimum weight exactly when it arrives earliest. By the way E is built, paths in G and feasible journeys are the same objects: a ride arc never lets the rider off between two consecutive stops, a change of trip identifier always passes through a transfer arc costing θ_{ch}, a wait arc lets the rider stay put, and no walk arc is longer than $\bar{t}_W$. A minimum-weight path is thus a feasible journey with the earliest possible arrival. With nonnegative weights, Dijkstra's (1959) algorithm is label-setting (Cormen et al., 2022). When the vertex with the smallest tentative label leaves the queue, nothing still queued has a smaller label, and since no arc weight is negative, no path found later can improve it. Each label is therefore final once its vertex is removed, and the first vertex removed at stop d carries the label $\tau^*(o, d, t_0)$. ∎

Remark 1 (Why the event-based vertex set matters). Proposition 1 rests on the event-based vertex set of Definition 4, which keeps every arrival and departure event at a stop as a separate vertex. A common shortcut in transit routing keeps only one label per stop, and that breaks the

proposition. Suppose Trip X reaches stop *s* at 07:20 and Trip Y reaches it at 07:22, Trip Y leaves *s* at 07:23, and $\theta_{ch} = 4$ min. A single-label search keeps only the 07:20 arrival. A passenger who came on Trip X cannot board before 07:24 because of the change time, misses Trip Y at 07:23, and has to wait for a later vehicle. The discarded 07:22 arrival, however, belongs to a passenger already on Trip Y, who simply stays on board and reaches the destination first. Section 3.5 measures how often this error occurs.

Proposition 2 (Computational complexity; Cormen et al., 2022). *Let $|S|$ be the number of stops, $|C|$ the number of connections, and $\bar{f}$ the mean number of footpaths per stop. Then $|V| = O(|C| + |S|)$ and $|E| = O(|C| + |S|\,\bar{f})$, and Proposition 1 can be implemented with a binary heap in $O(|E| \log |V|)$ time per query. If every arc duration is strictly positive, G is acyclic, and the same query can be answered in $O(|V| + |E|)$ time by a topological scan.*

Proof. Each connection contributes one departure vertex, one arrival vertex, and one ride arc. The wait arcs at a stop form a chain through its events and add O(1) arcs per event. Transfer and walk arcs add $O(\bar{f})$ for each arrival event. The heap bound is the standard bound for Dijkstra's algorithm with nonnegative weights (Dijkstra, 1959). In the acyclic case, relaxing the arcs in topological order visits each arc once; because every duration is strictly positive, every arc strictly increases the time coordinate, and no directed cycle can exist. ∎

Proposition 3 (Nested vehicle constraints and monotone reachability). *If $\gamma \le \gamma_0$, then $E_{\gamma 0} \subseteq E_{\gamma}$, and the shortest-path distances satisfy $d_{\gamma}(u, v) \le d_{\gamma 0}(u, v)$ for all $u, v \in V_R$. In particular, every vertex reachable under γ_0 is also reachable under γ.*

Proof. Take any arc $e \in E_{\gamma 0}$. In every component *i*, the arc limit satisfies $a^i_e \ge \gamma^i_0 \ge \gamma^i$, so $e \in E_{\gamma}$, and therefore $E_{\gamma 0} \subseteq E_{\gamma}$. Every path of $G_{\gamma 0}$ uses only arcs of $E_{\gamma 0}$ and is therefore also a path of G_{γ}. A minimum taken over a larger set of paths cannot exceed the minimum over a subset of them, so $d_{\gamma} \le d_{\gamma 0}$, and a finite $d_{\gamma 0}$ implies a finite d_{γ}, which gives the reachability claim. The argument needs only nonnegative arc weights and so applies directly to the travel-time weights ℓ_e/v_e. ∎

Corollary 4 (Vehicle-class accessibility ordering). *For every anchor a, type k, and budget θ, if $\gamma \le \gamma_0$ then $A_{a,\gamma}(k, \theta) \ge A_{a,\gamma 0}(k, \theta)$. A larger vehicle class never has greater accessibility than a smaller one.*

Proof. With travel times as weights, Proposition 3 makes every travel time under γ_0 at least as long as the corresponding time under γ. The indicator $1[\bar{T} \le \theta]$ is non-increasing in $\bar{T}$, so a sum of such indicators with nonnegative weights w_p is non-increasing as well. ∎

Remark 2 (Preprocessing many vehicle classes). Corollary 4 also keeps multi-class routing cheap. Under ≤ the vehicle classes form a partially ordered set, not a single chain: a tricycle needs more clearance than a sedan, while a sedan needs more width. Preprocessing therefore takes the classes in the order of a linear extension, most restrictive first. When class γ follows a comparable class γ_0 with $\gamma \le \gamma_0$, the inclusion $E_{\gamma 0} \subseteq E_\gamma$ lets the search for γ start from the reachable set of γ_0 and examine only the added arcs $E_\gamma \setminus E_{\gamma 0}$. Along a totally ordered chain (ten-wheel truck ≥ six-wheel truck ≥ four-wheel truck ≥ jeepney ≥ ambulance ≥ SUV ≥ pedestrian) this reduces to a single scan. Incomparable classes are processed separately.

Proposition 5 (Standard properties of the accessibility Gini coefficient; Cowell, 2011). *Let $A \ge 0$ with positive weights P and weighted mean $\mu > 0$. The Gini coefficient $G(A, P)$ satisfies (i) $0 \le G(A, P) \le 1$; (ii) scale invariance, $G(\lambda A, P) = G(A, P)$ for every $\lambda > 0$; and (iii) reduction under equal addition, $G(A + c1, P) < G(A, P)$ for every $c > 0$ whenever $G(A, P) > 0$.*

Proof. (i) Every term of the numerator is nonnegative, so $G \ge 0$. Since $A_a \ge 0$, $|A_i - A_j| \le A_i + A_j$, hence $\Sigma_i\Sigma_j P_iP_j|A_i - A_j| \le \Sigma_i\Sigma_j P_iP_j(A_i + A_j) = 2W\Sigma_i P_iA_i = 2W^2\mu$, and dividing by $2W^2\mu$ gives $G \le 1$. (ii) Multiplying A by λ multiplies both the pairwise differences and the mean μ by λ, and the factor cancels. (iii) Adding c to every score leaves all pairwise differences unchanged but raises the mean from μ to $\mu + c$, so G strictly decreases when $G > 0$. ∎

Remark 3 (What scale invariance means for policy). Property (ii) allows Gini values to be compared across destination types counted in different units, such as beds and classrooms, because the unit cancels. Property (iii) has a direct policy reading: raising access by the same absolute amount at every anchor lowers measured inequality.

Proposition 6 (Exact decomposition of the Theil index; Theil, 1967; Shorrocks, 1980). *Partition the anchors into groups g (here, municipalities). The Theil index then splits exactly into a within-group part and a between-group part. Let $W_g = \Sigma_{a \in g} P_a$, let μ_g be the group mean, and let $s_g = W_g\mu_g/(W\mu)$ be the group's share of all reachable destinations. Then*

$$T(A, P) = \underbrace{\sum_g s_g T_g}_{\text{within groups}} + \underbrace{\sum_g s_g \ln(\mu_g/\mu)}_{\text{between groups}} \tag{20}$$

Source: Theil (1967); decomposition after Shorrocks (1980)

where T_g is the Theil index within group g.

Proof. Write $p_a = P_a/W$ and, for $a \in g$, $p^g_a = P_a/W_g$, so that $p_a = (W_g/W)p^g_a$. Split the ratio as $A_a/\mu = (A_a/\mu_g)(\mu_g/\mu)$. Then

$$T = \sum_g s_g \Big[\sum_{a \in g} p^g_a \frac{A_a}{\mu_g} \ln \frac{A_a}{\mu_g} + \ln \frac{\mu_g}{\mu} \sum_{a \in g} p^g_a \frac{A_a}{\mu_g} \Big] \tag{21}$$

Source: expansion of Equation (20), following Theil (1967) and Shorrocks (1980)

Inside the brackets, the first sum is T_g by definition and the second is $\Sigma_{a \in g}\ P_a A_a/(W_g \mu_g) = 1$. Substituting both gives Equation (20). A group with $\mu_g = 0$ has $s_g = 0$ and adds nothing to either part. ∎

Remark 4. Proposition 6 turns a single inequality figure into a diagnosis. A dominant within-municipality term points to a local problem, to be met with feeder routes and better stop placement inside the municipality. A dominant between-municipality term points to a structural problem, to be met with trunk service linking municipalities. The Gini coefficient has no exact split of this kind (Cowell, 2011; Shorrocks, 1980), so the two indices are reported side by side.

Remark 4a (Decomposition over a chosen index set). Q4 raised the case of an index set made of landmarks instead of a full partition of the province. The proof of the decomposition does not depend on spatial completeness, so it remains valid: all it needs is a finite index set, positive weights, and a partition, and the partition here is given by the municipality of each anchor. What changes is the interpretation, not the algebra. Each municipality is represented by its anchors, and a municipality with only one anchor enters the decomposition as a single point instead of an average over several places. This is the main reason Section 3.6.1 varies the anchor set.

Proposition 7 (Monotonicity and subadditivity of regulated fares). *The leg fare $f(d) = f_0 + r\lceil \max(0, d - d_0) \rceil$ is nondecreasing in d. If, in addition, $rd_0 \le f_0$ and d_0 is a positive integer, then f is subadditive: $f(d_1 + d_2) \le f(d_1) + f(d_2)$ for all $d_1, d_2 \ge 0$.*

Proof. Monotonicity is immediate, since $d \mapsto \lceil \max(0, d - d_0) \rceil$ never decreases and $r > 0$. For subadditivity, put $x = \max(0, d_1 - d_0)$ and $y = \max(0, d_2 - d_0)$. From $d_1 \le x + d_0$ and $d_2 \le y + d_0$ it follows that $\max(0, d_1 + d_2 - d_0) \le x + y + d_0$. The ceiling of a sum never exceeds the sum of the ceilings, and the integer d_0 is its own ceiling, so $\lceil \max(0, d_1 + d_2 - d_0) \rceil \le \lceil x \rceil + \lceil y \rceil + d_0$. Hence

$$f(d_1 + d_2) \le f_0 + r\lceil x \rceil + r\lceil y \rceil + rd_0 \le 2f_0 + r\lceil x \rceil + r\lceil y \rceil = f(d_1) + f(d_2) \tag{22}$$

Source: formulated in this study (Proposition 7)

where the second step uses $rd_0 \le f_0$. ∎

Corollary 8 (Fare and time can conflict). *For the traditional jeepney of Table 4, $rd_0 = 8 \le 14 = f_0$, so its fare is subadditive: one ride over a given distance never costs more than two rides covering the same distance.* The cheapest journey therefore favors a few long legs, while the fastest may involve more transfers. Because the two aims can pull apart, the planner returns the whole Pareto set over (arrival time, fare, number of transfers) instead of a single journey.

Remark 5. Finding shortest paths under two or more additive criteria is hard in general. The weight-constrained version is already NP-complete (Garey & Johnson, 1979), and the number of Pareto-optimal paths can grow exponentially with the size of the network (P. Hansen, 1980). Here, however, fares take only a few distinct integer values and no journey boards more than four times, so a label-correcting search that prunes by Pareto dominance (Martins, 1984) enumerates the frontier quickly. The same practical reasoning underlies the multi-criteria version of RAPTOR (Delling et al., 2015).

The remaining results, Propositions 9, 10, and 12 and Corollary 11, concern the hospital extension (Sections 2.7 to 2.9).

Proposition 9 (Early stopping in hospital mode loses no exactness). *Search from o at time t, by the label-setting method of Proposition 1 in mode pt or by Dijkstra's algorithm on $G_{\gamma amb}$ in mode amb, and stop as soon as the vertex of a suitable hospital has been extracted together with every other vertex that has the same label. Then $h^*(o, \pi, t)$ is the hospital chosen by the tie-break of Definition 17, its label gives $T^*(o, \pi, t)$, and no vertex with a larger label is ever extracted. If no suitable hospital can be reached, the queue empties and $T^* = +\infty$.* The rule is Dijkstra's (1959) single-target stopping condition applied to the whole set of suitable hospitals.

Proof. Both graphs have nonnegative arc weights (G by Assumption 1, and $G_{\gamma amb}$ because lengths and speeds are positive), so labels leave the queue in nondecreasing order and are final, exactly as in Proposition 1. Let L be the label at which a suitable hospital vertex first leaves the queue. Every other suitable hospital vertex has a label of at least L, since a smaller label would have been extracted earlier. Because the search continues until every label left in the queue exceeds L, every suitable hospital vertex with label exactly L is extracted before the search stops. These vertices are precisely the minimizers in Equation (16), and the tie-break selects h^* among them. In public transport mode, the sink z_h can be entered only through walking arcs from arrival events, so its label is the earliest possible arrival at the facility and $T^* = L - t$. No vertex with a label above L ever leaves the queue. If no facility in $H_\pi(t)$ is reachable, the stopping condition never fires and the queue simply empties. ∎

Remark 6 (Why a target set saves work). Hospital mode uses the same transit graph but asks a different question. An accessibility query has to find paths from an origin to every landmark, whereas a hospital query needs only the single nearest suitable facility, so the search can stop early and do far less work. Proposition 9 confines the search to vertices nearer than the nearest suitable hospital, which on a large network is a small share of the whole; Table 14 measures that share.

Proposition 10 (Suitability is monotone in ability to pay and in acuity). *Fix o, t, the mode of travel, and every bed status. (i) For $\rho \le \rho'$, $H_{(\kappa,\rho)}(t) \subseteq H_{(\kappa,\rho')}(t)$ and $T^*(o, (\kappa, \rho'), t) \le T^*(o, (\kappa, \rho), t)$. (ii) If κ and κ' are both inside K_E or both outside it, and $\lambda(\kappa) \ge \lambda(\kappa')$, then $H_{(\kappa,\rho)}(t) \subseteq H_{(\kappa',\rho)}(t)$ and $T^*(o, (\kappa', \rho), t) \le T^*(o, (\kappa, \rho), t)$. (iii) If $\kappa \in K_E$, then $H_{(\kappa,\rho)}(t) = H_{(\kappa,\rho max)}(t)$ whatever ρ is.*

Proof. (i) Of the four factors in Equation (15), only the last involves ρ, and $1[\kappa \in K_E \text{ or } q_h \le \rho]$ is nondecreasing in ρ. Hence $\Phi(h; (\kappa, \rho), t) \le \Phi(h; (\kappa, \rho'), t)$ for every h, and the suitable sets are nested. As in Proposition 3, a minimum over a set is never below the minimum over a larger set containing it. (ii) When κ and κ′ fall on the same side of K_E, the second and fourth factors agree for both profiles and the third does not depend on κ, while $\lambda(\kappa) \ge \lambda(\kappa')$ gives $1[\eta_h \ge \lambda(\kappa)] \le 1[\eta_h \ge \lambda(\kappa')]$. The sets are again nested, and the minimum travel times compare as in part (i). (iii) With $\kappa \in K_E$, the fourth factor equals 1 whatever ρ is. ∎

Remark 7 (A deliberate break at the emergency boundary). Profile pairs on opposite sides of K_E have to be left out of Proposition 10(ii). Moving from κ = 3 to κ = 2 tightens the level requirement, which removes hospitals, but it also switches off the payment filter, which adds them. At anchor A6 of instance S1 the additions win. With ρ = 0, the urgent patient (κ = 3) is routed 37.2 min to S4, whereas the emergent patient (κ = 2) is sent to S7, only 13.0 min away (Table 21). The sicker patient ends up closer to care because the law removes the payment barrier, which is exactly the intended behavior.

Corollary 11 (The payment gap). *For all o, t, κ, and ρ: $\Delta(o, \kappa, \rho, t) \ge 0$; Δ is non-increasing in ρ; $\Delta(o, \kappa, \rho_{max}, t) = 0$; and $\Delta(o, \kappa, \rho, t) = 0$ for every $\kappa \in K_E$.*

Proof. The first three statements follow from Proposition 10(i). If $T^*(o, (\kappa, \rho_{max}), t) = +\infty$, then even dropping the payment filter leaves no suitable hospital within reach; $T^*(o, (\kappa, \rho), t)$ is then infinite as well, and Equation (18) sets Δ = 0 by convention. If only $T^*(o, (\kappa, \rho), t)$ is infinite, then $\Delta = +\infty$, a case reported separately as "no affordable suitable hospital reachable." The last statement follows from Proposition 10(iii). ∎

Remark 8 (Reading the gap and the bed vectors). Outside emergencies, the payment gap counts the minutes that a limited budget adds to a trip; for emergency patients it is zero by construction. Taken over all anchors, the gap and the free-bed scores B_a and B^{ρ}_a of Equation (19) form

nonnegative vectors. Lorenz curves, Gini and Theil indices, and Palma ratios (Section 2.4) can therefore be computed on them directly, and Propositions 5 and 6 hold whenever the mean is positive.

Remark 9 (The ambulance's place in the vehicle order). In every component of Table 3, the ambulance lies between the sport utility vehicle and the jeepney. By Proposition 3, every road open to a jeepney-class vehicle is therefore open to the ambulance, and with equal arc speeds Corollary 4 guarantees that its hospital accessibility is at least that of a jeepney. Where a posted width limit or bridge load limit closes a road to the ambulance, hospital mode simply routes around it.

Proposition 12 (Bounds on messages, opt-outs, and report age). *Assume the protocol of Definition 13. (i) No hospital receives more than $1 + n_{\mathrm{rem}}$ messages in a cycle, or more than $(1 + n_{\mathrm{rem}})\lceil 24\ \mathrm{h}/P_{\mathrm{h}} \rceil$ messages in a day. (ii) Nothing is sent to a hospital once its opt-out is in the project mailbox. (iii) A hospital that answers every inquiry within D_{rep} with a reply that passes validation has $age_{\mathrm{h}}(t) < P_{\mathrm{h}} + D_{\mathrm{rep}}$ at all times after its first such reply, so it is never UNKNOWN if $P_{\mathrm{h}} + D_{\mathrm{rep}} \le age_{\max}$.*

Proof. (i) Within a cycle, the Coordinator sends one inquiry at t_{k} and reminders only at $t_{\mathrm{k}} + jR_{\mathrm{rem}}$ with $j \le n_{\mathrm{rem}}$, and the condition $n_{\mathrm{rem}}R_{\mathrm{rem}} < P_{\mathrm{h}}$ keeps every reminder inside its own cycle. (ii) The Reader works through the mailbox immediately before each send, and the Coordinator consults the opt-out list only afterwards, so an opt-out already in the mailbox when a send falls due is recorded before that send. (iii) Let $d_{\mathrm{k}} \le D_{\mathrm{rep}}$ be the time the hospital took to answer cycle k. For $t \ge t_1 + d_1$, choose the largest k with $t_{\mathrm{k}} + d_{\mathrm{k}} \le t$. Then $t < t_{\mathrm{k}+1} + d_{\mathrm{k}+1} \le t_{\mathrm{k}} + P_{\mathrm{h}} + D_{\mathrm{rep}}$, and since rule V5 keeps the most recent report, the report on file dates from $t_{\mathrm{k}} + d_{\mathrm{k}}$ or later. Thus

$$age_{\mathrm{h}}(t) \le t - (t_{\mathrm{k}} + d_{\mathrm{k}}) < P_{\mathrm{h}} + D_{\mathrm{rep}} - d_{\mathrm{k}} \le P_{\mathrm{h}} + D_{\mathrm{rep}} \tag{23}$$

Source: formulated in this study (Proposition 12)

which proves the claim. ∎

With the default parameters of Definition 13 ($P_{\mathrm{h}} = 8$ h and $n_{\mathrm{rem}} = 1$) and replies that arrive within $D_{\mathrm{rep}} = 2$ h, Proposition 12 allows at most 2 messages per cycle and 6 per day and keeps every report younger than 10 h, well inside the 12 h limit of Definition 14. On a daily schedule ($P_{\mathrm{h}} = 24$ h), a hospital receives at most 2 messages a day, but the age bound grows to 26 h, so its status would lapse to UNKNOWN between reports unless $age_{\max}$ is raised.

Remark 10 (Scope of Proposition 12). Proposition 12 bounds how often messages are sent and how stale the reports of responsive hospitals can become. It promises nothing about response rates, since replying is voluntary. DataCollect reached 98 percent reporting with strong national backing

(Cabaro et al., 2023); this project has no such backing and makes no claim about how many hospitals would respond.

3.3 Software Verification

To keep programming errors out of the analysis, every claimed property was checked numerically against a separate reference implementation. Table 11 summarizes the checks on the transport engine, and Table 12 those on hospital mode and the agent network. All of these figures come from synthetic test cases, not from field data.

Row 2 of Table 11 deserves attention. With limited development time, the single-pass Connection Scan Algorithm (CSA) is an obvious choice for the engine. As Dibbelt et al. (2018) point out, however, a single scan requires footpaths that are transitively closed, and when they are not, it fails without warning. In this test it returned a late arrival time for almost half of the reachable queries (47.5 percent), late by 32.5 minutes on average, and raised no error or warning of any kind.

An atlas built on such an engine would understate access almost everywhere and would mark well-served landmarks as underserved. Every computation reported here therefore uses the verified label-setting engine.

Table 11. Pipeline checks, all produced by the project's test suite on random synthetic instances whose correct answers were obtained independently.

Property Checked	Method of Check	Result
Correctness of earliest arrival (Prop. 1)	618 random queries over 120 synthetic timetables, checked against complete enumeration of feasible journeys	577 reachable; arrival times agree in every case
Naive single-scan implementation	The same queries through a one-pass scan whose footpaths are not transitively closed	Late in 274 of 577 reachable queries (47.5%); mean error 32.5 min; worst 109 min
Gini scale invariance (Prop. 5 ii)	500 random weighted vectors, λ drawn from [0.2, 12]	Largest deviation 2.2×10^{-16}
Gini under translation (Prop. 5 iii)	500 random weighted vectors, *c* drawn from [0.5, 20]	Lower index in all 500 cases
Theil decomposition (Prop. 6)	500 random two-group partitions	Largest residual 5.6×10^{-16}
Vehicle-class monotonicity (Prop. 3)	300 random road graphs; restricted class made by deleting 35% of the arcs	Zero violations over 11,620 origin–destination pairs
Fare monotonicity and subadditivity (Prop. 7)	Distance sweep to 60 km and 1,000 random leg splits	Monotone over the sweep; subadditive in every split

Hospital mode and the agent network were checked in the same way by verify_hospital_mode.py, run with a fixed random seed. The ambulance router was compared with two independent references: all-pairs distances from the Floyd–Warshall algorithm (Floyd, 1962; Warshall, 1962), which share no code with the priority-queue engine, and a rewrite of the suitability rule as set operations. The public transport router was compared with a dynamic program that sweeps the time-expanded graph in topological order (Cormen et al., 2022), which works because every arc moves forward in time (Proposition 2). For the agents, 60 synthetic hospitals were simulated over 30 days, or 90 inquiry cycles. Nine never replied and six sent noisy replies; six more opted out after two to ten messages; fifteen replied irregularly and in varied wording; and the remaining twenty-four answered every inquiry within two hours on the reply form. On top of this, 200 forged replies were planted in the mailbox.

Table 12. Checks on hospital mode and on the agent network (synthetic inputs only; no real hospital or mailbox).

Property Checked	Method of Check	Result
Early stopping, ambulance mode (Prop. 9)	200 random road graphs (120–260 vertices, 6–15 hospitals, roughly 23% of arcs closed to the ambulance class); 4,000 random origin–profile queries against Floyd–Warshall distances and a set-based suitability filter	Reachable suitable hospital in 2,911 queries; hospital and travel time agree in all
Early stopping, patient mode (Prop. 9)	120 random timetables (12–20 stops, 4–6 routes, 3–6 hospitals with sink vertices); 1,200 queries against a dynamic program in topological order	Reachable suitable hospital in 660 queries; full agreement; no arc runs backward in time
Nesting and monotonicity (Prop. 10)	Each of the 20 profiles at every ambulance query: 140,000 set-inclusion checks and 140,000 travel-time checks; 25,200 travel-time checks on the transit queries	No violation found
Payment gap (Cor. 11)	80,000 origin–profile pairs drawn from the ambulance instances	Never below zero; zero for every emergency profile
Reader extraction (rule V2)	2,400 synthetic replies from 12 templates: 8 readable (reply form; English or Filipino free text with dates and telephone numbers as distractors) and 4 unreadable or contradictory	1,600 of 1,600 readable replies read exactly; 800 of 800 others referred to review; no wrong value
Message load (Prop. 12 i)	Mock mail environment, 60 hospitals, 30 days; 7,225 messages sent (4,902 inquiries and 2,323 reminders)	No hospital got more than 2 messages per cycle or 6 per day
Opt-out (Prop. 12 ii)	6 hospitals opting out after 2 to 10 messages	Nothing sent after an opt-out arrived

Property Checked	Method of Check	Result
Report age (Prop. 12 iii)	24 hospitals replying within 2 h; age logged every 10 min	Oldest 579.8 min, inside the 600 min bound; never UNKNOWN after the first report
Validation (Table 8)	3,849 replies from the 51 responding hospitals, 67 of them with capacity typed ten times too large	3,302 accepted, none carrying a wrong value; 547 held, all 67 typing errors among them
Authentication (rule V1)	200 forged replies: never-issued tokens, genuine tokens from a look-alike address, or another hospital's token	None accepted

Two rows of Table 12 call for comment. The Reader row is the weakest. Its twelve templates were written together with the extraction rules, so the row shows only that the rules work as designed, refusal to guess included. Real staff may word their replies in ways no template anticipated, which is why an unreadable reply always goes to a person (Limitation L13). The report-age row is the strongest. Among the prompt responders, the oldest report reached 579.8 min, within the 600 min bound of Proposition 12, and the 12 h cutoff of Definition 14 never had to intervene.

3.4 Computational Cost

Table 13. Mean time per query for the verified engine; each configuration averages 40 queries.

Stops	Connections	Mean Query Time (ms)
50	900	1.03
100	2,288	1.41
200	5,620	3.66
400	12,576	15.06
800	28,336	69.51

A fit of log(time) against log $|C|$ has a slope of 1.243 with $R^2 = 0.933$, consistent with the near-linear growth of Proposition 2 once heap overhead is counted. In the largest case, about 28,000 connections, one query took roughly 70 ms in plain, unoptimized Python on a single core. A laptop could therefore complete the full anchor-to-destination sweep for the study area within a few hours, with no server. Working from landmarks lightens the load further, since there are far fewer anchors than the 720 barangays.

Because its search stops early (Proposition 9), hospital mode was timed separately. Table 14 compares it with the one-to-all search that the atlas uses. The test graphs were random road graphs

of increasing size with one hospital for every 150 vertices; patient profiles were drawn at random, and origins were taken from the largest connected component of the ambulance subgraph.

Table 14. Early-stopping hospital search compared with a full search (random road graphs, 60 queries per size, plain Python on one core).

Vertices	Arcs Open to the Ambulance	Hospitals	Early-Stopping Query (ms)	Full Search (ms)	Share of Vertices Settled
1,000	3,768	8	0.34	0.68	44.1%
4,000	14,898	26	0.61	3.58	13.8%
16,000	59,926	106	1.33	22.07	5.2%

On the largest graph, with 16,000 vertices, the full search settled about twenty times as many vertices as the early-stopping search and took about seventeen times as long. The advantage grows with the size of the network, because the nearest suitable hospital does not move farther away as the graph grows. In practice, this speed allows the route to be recomputed whenever the ambulance moves or a new bed report arrives, with answers in milliseconds even on a province-wide road network.

3.5 Worked Instance S1

Instance S1 is a small synthetic test network, built so that every stage of the pipeline can be traced by hand and checked against independently computed answers. It does not represent Albay, and this subsection makes no empirical claim about any real place.

Table 15. Instance S1 in brief.

Element	Value
Stops	7 (S1–S7)
Routes	4 base routes, run in both directions
Connections in the service day	787
Minimum change time θ_{ch}	4 min
Walking cap $\bar{t}_W$	15 min
Footpaths	S3–S6 (11 min), S2–S5 (13 min)
Departure grid T_0	06:00–08:00 every 5 min (25 departure times)
Primary baseline budget θ	30 min
Destination layer	hospital beds at S4 (120), S7 (40), and S3 (25)
Anchors	six, A1–A6, split between municipalities M1 and M2

Table 16. Mean travel time from door to door (min) over the departure grid, from each anchor of S1 to each hospital site, with the resulting 30-minute accessibility in reachable beds (synthetic instance).

Anchor	Weight	Municipality	to S4 (120)	to S7 (40)	to S3 (25)	$A_a(30)$
A1	4,200	M1	29.2	49.0	24.6	145
A2	3,100	M1	27.3	61.0	21.3	145
A3	1,850	M1	18.6	51.4	7.0	145
A4	1,400	M2	43.6	76.6	37.5	0
A5	980	M2	32.6	65.8	26.0	25
A6	2,600	M2	37.2	13.0	48.1	40

Table 17. Inequality of access to hospital beds in S1 at the 30-minute baseline (synthetic instance).

Quantity	Value
Weighted mean μ	102.99 beds
Gini coefficient G	0.2747
Theil index T	0.2205
Part within municipalities	0.0308 (14.0% of total)
Part between municipalities	0.1897 (86.0% of total)
Identity check $\lvert T - (\text{within} + \text{between})\rvert$	2.8×10^{-17}
Palma ratio	0.907
Weight share with zero access	9.9%

Anchor A1, 29.2 minutes from the 120-bed facility, falls inside the budget, while anchor A5, at 32.6 minutes, falls just outside it. That difference of 3.4 minutes cuts the score from 145 to 25. The jump is not a quirk of S1; it is built into every cumulative measure with a hard threshold. This is why Section 3.6 sweeps the travel budget and why the smooth decay measure of Definition 7 is part of the framework.

The decomposition behaves as intended. The difference between the two municipalities accounts for 86 percent of total inequality, which points to weak trunk links between municipalities, more than to poor feeder service within them, as the main cause. Whether the full Albay instance shows the same pattern is a question S1 cannot answer; Hypothesis M3 (Section 3.6.3) tests it.

Table 18. Leg fares under the traditional jeepney parameters of Table 4.

Leg Length *d* (km)	Full Fare (₱)	Fare with 20% Discount (₱)
1.0	14.00	11.20
4.0	14.00	11.20
4.2	16.00	12.80
7.0	20.00	16.00
10.5	28.00	22.40
18.0	42.00	33.60

Table 19. Independent tests on S1 and on new random instances. The reference in row 1 is a depth-limited exhaustive enumeration, written as plain recursion, that shares no logic with the priority-queue engine under test.

Property	Method	Result
Correctness of earliest arrival (Prop. 1)	400 random queries on S1 against exhaustive enumeration	400 reachable; full agreement
Labeling by stop only (Remark 1)	The same 400 queries with one label kept per stop	Wrong in 6 of 400 (1.5%); mean error 15.0 min; worst 30 min
Gini scale invariance (Prop. 5 ii)	500 random weighted vectors, λ in [0.2, 12]	Largest deviation 1.1×10^{-16}
Gini under translation (Prop. 5 iii)	500 random weighted vectors, *c* in [0.5, 20]	Lower in all 500
Theil decomposition (Prop. 6)	500 random three-group partitions	Largest residual 1.7×10^{-16}
Vehicle-class monotonicity (Prop. 3)	60 random road graphs, width threshold raised from 1.3 m to 2.2 m	Zero violations in 5,400 origin–destination pairs
Fare monotonicity and subadditivity (Prop. 7)	Sweep from 0 to 60 km in 0.1 km steps and 1,000 random leg splits	Monotone throughout; subadditive in every split

Row 2 shows the second implementation pitfall that this study uncovered, and it is much harder to spot than the first. The single-scan defect of Table 11 reveals itself quickly, since it affects nearly half of all queries with delays that average about half an hour. Stop-only labeling fails rarely, and each failure costs at most one headway, so looking over a few sample journeys is unlikely to reveal anything wrong. The flaw comes to light only when the outputs are compared systematically with an independently written reference. The lesson is to build a ground-truth reference routine before building a fast routing engine.

3.5.1 Extension: Hospital Mode on Instance S1

To run hospital mode end to end, Table 20 assigns invented attributes to the three hospital sites of S1. These attributes are the only new input, since the travel times come from Table 16. S1 has no ambulance road layer, so all trips use public transport (x = pt), and all reports are treated as fresh. Tables 20 to 22 were computed by verify_hospital_mode.py, whose inequality routine reproduces Table 17 to the fourth decimal place.

Table 20. Synthetic hospital attributes created for S1. Every site has an emergency room (e_h = 1).

Site	c_h (beds)	η_h	Ownership and Programs	q_h	$\hat{o}_h$	b_h	σ_h
S4	120	3	Government; with Malasakit Center	0	112	8	AVAILABLE
S7	40	2	Private; accredited by PhilHealth	2	31	9	AVAILABLE
S3	25	1	Government; without Malasakit Center	1	25	0	FULL

Table 21. Recommendation and travel time (min) for each anchor of S1 under several patient profiles. The Plain Nearest column ignores level, beds, and cost (synthetic instance).

Anchor	Plain Nearest	κ = 1, any ρ	κ = 2, ρ = 0	κ = 3, ρ = 0	κ = 3, ρ = 3	κ = 5, ρ = 1
A1	S3, 24.6 (full)	S4, 29.2	S4, 29.2	S4, 29.2	S4, 29.2	S4, 29.2
A2	S3, 21.3 (full)	S4, 27.3	S4, 27.3	S4, 27.3	S4, 27.3	S4, 27.3
A3	S3, 7.0 (full)	S4, 18.6	S4, 18.6	S4, 18.6	S4, 18.6	S4, 18.6
A4	S3, 37.5 (full)	S4, 43.6	S4, 43.6	S4, 43.6	S4, 43.6	S4, 43.6
A5	S3, 26.0 (full)	S4, 32.6	S4, 32.6	S4, 32.6	S4, 32.6	S4, 32.6
A6	S7, 13.0	S7, 13.0 (stabilize), then S4, 37.2	S7, 13.0	S4, 37.2	S7, 13.0	S4, 37.2

Table 21 shows four effects. The first is bed availability. For five of the six anchors the physically nearest hospital is S3, which has no free beds at the moment. Hospital mode sends these patients to S4 instead, adding between 4.6 and 11.6 minutes of travel; the largest detour is at anchor A3, where the trip grows from 7.0 to 18.6 minutes. A router that ignored bed counts would have sent them to a hospital that could not admit them.

The second effect is capability. Even if it had free beds, S3 could not provide the Level 2 or higher care that the more urgent profiles require.

The third effect is ability to pay. At anchor A6, an urgent patient ($\kappa = 3$) who needs free or assisted care ($\rho = 0$) is routed 37.2 minutes to S4, while a patient able to pay for private care ($\rho = 3$) is routed 13.0 minutes to S7, a payment gap of 24.2 minutes. Weighted over all six anchors, the mean gap for this profile is 4.45 minutes, all of it coming from anchor A6, which carries 18.4 percent of the weight. Read against SDG target 3.8, these 24.2 extra minutes are the travel penalty that a lack of financial protection imposes, and the payment-gap measure turns that penalty into a number that can be mapped.

The fourth effect comes from the two emergency rules. For $\kappa = 2$ the gap is zero at every anchor, as Corollary 11 requires. When the script was run a second time with the payment override switched off, the 24.2 min gap reappeared at A6 for an emergency patient, which confirms that the override is what removes it. For $\kappa = 1$, S4 is the only Level 3 site, so a patient at A6 must end up 37.2 min away, yet the emergency room of S7 is only 13.0 min off. Since $13.0 + 15 < 37.2$, the patient is first stabilized at S7. No other anchor triggers the rule, because S4 is already the nearest open emergency room for each of them.

Table 22. Beds reachable within 30 min in S1, counted three ways: authorized, free, and free in hospitals that payment class 0 accepts (synthetic instance).

Quantity	**Authorized, $A_a(30)$**	**Free, $B_a(30)$**	**Free and Accepted, $B^0_a(30)$**
Scores, A1 to A6	145, 145, 145, 0, 25, 40	8, 8, 8, 0, 0, 9	8, 8, 8, 0, 0, 0
Weighted mean μ	102.99	6.84	5.18
Gini coefficient	0.2747	0.1859	0.3524
Theil index	0.2205	0.1857	0.4345
Share within municipalities	14.0%	84.8%	0.0%
Share between municipalities	86.0%	15.2%	100.0%
Palma ratio	0.907	0.486	2.103
Weight share with zero access	9.9%	16.8%	35.2%

Table 22 shows how the inequality diagnosis changes when free beds replace authorized capacity. For authorized beds, 86 percent of the Theil index comes from differences between municipalities, the trunk-network deficit noted earlier. For free beds, 84.8 percent of the inequality moves inside M2: anchor A6 reaches the nine free beds at S7, while anchors A4 and A5 reach none. Counting only hospitals that accept payment class 0 moves the inequality back between municipalities, because S4 alone qualifies and only the anchors of M1 reach it within 30 minutes. These figures come from a synthetic seven-stop network with invented bed counts, yet the methodological point

holds: counting live beds can change which policy the decomposition recommends, so the three vectors should always be reported together.

3.6 Computational Experiments on the Open-Data Instance

These experiments run on the dataset that the pipeline of Section 3.1 produces for the study area, and on randomized variants of it. They examine how the system behaves and whether its outputs hold up under different parameter settings. Whether those outputs match actual operations in Albay is a separate question that needs field survey data, and Section 4.3 leaves it to future work.

3.6.1 Sensitivity Analysis

Sensitivity testing varies the anchor sets, headway values, vehicle-class limits, departure-time assumptions, and travel budgets. The main baseline is 30 minutes, and the other budgets show whether a conclusion depends on the threshold chosen. Since the choice of landmarks is a key modeling decision, the anchor set is varied on purpose, with three configurations: terminals only, terminals plus municipal halls, and all anchors of Table 2. Only directional conclusions that hold in all three configurations count as robust.

Hospital mode is re-run with two alternative acuity tables, one lowering every $\lambda(\kappa)$ by one level and the other requiring Level 3 care for both emergency levels, and with several stabilization margins θ_{stab} and freshness limits age_{max}. The open-data instance contains no live bed reports, so every facility sits at UNKNOWN (Definition 14) and remains eligible for routing. These runs therefore test capability and financial access, not live bed capacity.

3.6.2 Statistical Treatment

Eight questions are tested formally, six on the transport layer and two on hospital mode. Each question names an estimator and a decision rule at $\alpha = 0.05$, and Holm's (1979) step-down procedure is applied across the whole family of tests. Every test also reports an effect size, so that a difference too small to matter for planning is not presented as important. Because all tests run on synthetic model outputs, any significance they show describes the model, not actual road conditions.

Table 23. How each question is tested.

Question	Test or Estimator	Reason for the Choice
(S1) Is modeled access more unequal than resampling noise alone would make it?	Weighted Gini (Gini, 1912) with a 95% percentile bootstrap interval from 2,000 resamples (Efron, 1979; Efron & Tibshirani, 1993)	Assumes no distribution; the interval reflects the sampling noise that comes from the anchor set

Question	Test or Estimator	Reason for the Choice
(S2) Do urban and rural anchors differ?	Mann–Whitney U (Mann & Whitney, 1947), with the rank-biserial correlation as effect size (Kerby, 2014)	Access is skewed to the right and floored at zero, which rules out a t-test
(S3) Is the inequality local or structural?	Theil decomposition (Prop. 6; Shorrocks, 1980) with a bootstrap interval for the within-group share	An exact decomposition separates the two explanations cleanly
(S4) Does modeled access track settlement density?	Spearman rank correlation (Spearman, 1904)	No linear relation is assumed
(S5) How sensitive is the result to the anchor set?	Median discrete Fréchet distance between Lorenz curves (Eiter & Mannila, 1994)	Compares whole curves instead of single points
(S6) How sensitive is the result to the headway assumption?	Spread of results when headways are perturbed by ±25% and ±50%	Headway is the least constrained input of the model
(S7) Do urban and rural anchors differ in payment gap?	Mann–Whitney U with the rank-biserial effect size (Kerby, 2014)	The gap is nonnegative and has many zeros
(S8) How often is the plain nearest hospital unsuitable?	Weighted share of anchors with a 95% percentile bootstrap interval from 2,000 resamples (Efron & Tibshirani, 1993)	A proportion is better described by an interval than by a test

3.6.3 Pre-Registered Model Hypotheses

All hypotheses and decision rules below were fixed before the study-area instance was run. They are statements about the open-data model and can be decided without field measurements or hospital replies.

- **M1.** At the 30-minute baseline, modeled access to hospital beds has a weighted Gini coefficient above 0.30. M1 holds if the bootstrap interval lies entirely above 0.30.
- **M2.** Urban-center anchors have higher modeled accessibility than rural anchors, tested with the Mann–Whitney U at $\alpha = 0.05$.
- **M3.** The between-municipality term is less than half of the total, which would make modeled transport disadvantage mainly a matter of local feeder service.
- **M4.** A larger vehicle class never reaches more of the road network. Proposition 3 makes this exact, not merely typical, so a single violation would point to a faulty road attribute and send that segment back for checking.
- **M5.** Under every anchor set of Section 3.6.1 and every $\theta \in \{15, 30, 45\}$ min, hypotheses M1, M3, M6, and M7 point the same way.

- **M6.** With $\kappa = 3$ and $\rho = 0$ at the 30-minute baseline, the payment gap is larger at rural anchors than at urban ones, tested with the Mann–Whitney U at $\alpha = 0.05$.
- **M7.** With $\kappa = 2$, more than half of the anchor weight has a plain nearest hospital that fails the capability test of Definition 16. M7 is supported if the bootstrap confidence interval lies entirely above one half.
- **M8.** For both emergency levels ($\kappa \in K_E$), no anchor has a nonzero payment gap. Corollary 11 makes this exact, so any nonzero value would indicate a software or coding error.

Within the model, M2 and M6 measure urban–rural gaps in access that bear on SDG targets 10.2 and 11.2, and M7 measures how often the nearest hospital falls short of the care that target 3.8 assumes a patient can reach. Negative results will be reported as well. Fairly even modeled access, or a small payment gap, would be as useful to a planner as the opposite finding, and either will be reported without changing the analysis.

3.6.4 Robustness Sweep

Six sensitivity sweeps recompute the main results over ranges of parameters: travel budgets of 15, 30, and 45 min; departure windows in the morning peak, at midday, and in the evening; decay parameters for the measure of Definition 7, tuned to median half-lives of 15, 30, and 45 minutes; a half-headway waiting model; each of the three anchor sets; and, for hospital mode, the alternative acuity tables, freshness limits, and stabilization margins of Section 3.6.1. A conclusion counts as robust only if its direction holds in every sweep.

4. CONCLUSIONS

4.1 Assessment of Results

The primary question asked for a measure of accessibility attached to each anchor landmark and for a way to describe how that measure varies across space. The study's answer is that such a measure can be given a clear theoretical footing, computed exactly, and recomputed quickly whenever the network changes, all without field data. Four core results hold independently of any input specific to Albay.

- **Algorithmic correctness.** Label-setting search on the time-expanded graph returns exact earliest arrival times (Proposition 1). The implementation agreed with brute-force enumeration on all 1,018 test queries, 618 in Table 11 and 400 in Table 19.
- **Vehicle ordering.** By Proposition 3 and Corollary 4, the road networks of the vehicle classes are nested under a partial order. Accessibility therefore never increases with vehicle size, and a single linear extension of the order lets all ten classes, the ambulance included, be preprocessed in one pass.
- **Well-behaved inequality measures.** The Gini coefficient is scale-invariant, so it can compare inequality across beds, classrooms, and service windows, and the Theil index splits exactly into within-municipality and between-municipality parts, which separates feeder problems from trunk problems.
- **Fare subadditivity.** The current fare structure is subadditive (Proposition 7 and Corollary 8), so the router offers a Pareto frontier of trips instead of a single recommendation.

A fifth finding carries direct practical weight. Two common implementation shortcuts were tested against independently computed ground truth. A single-pass scan without transitively closed footpaths returned late arrivals for 47.5 percent of reachable queries, by 32.5 minutes on average. Keeping one label per stop returned late arrivals for 1.5 percent of queries, by 15 minutes on average. The rarity of the second error makes it the more dangerous of the two in practice. The maps produced by either shortcut look plausible, and only a systematic comparison with a reference implementation exposes them.

The sixth finding goes to the main goal of the study. Before this work there was no machine-readable model of public transport for the 18 LGUs of Albay, and therefore no way to check one. The study shows that open registers are enough to build such a model: OpenStreetMap supplies the roads, the national health and school registers supply the destinations, the census supplies the demand weights, and the LTFRB fare orders supply the fares. Field calibration would make the

model more accurate, and the pipeline is built so that field data can replace the generated feed without changes downstream (Section 4.3).

The hospital extension adds three further results.

- **Exact early-stopping router.** On a 16,000-vertex network the hospital search settles only about 5 percent of the vertices, and it matched the reference solution in all 5,200 test queries (Proposition 9; Tables 12 and 14).
- **Monotone suitability and the payment gap.** Widening the payment class can only add eligible hospitals, and raising the required service level can only remove them, so the payment gap is never negative. The emergency rule of Republic Act No. 10932 (2017), written into the model as an indicator function, sets the gap to zero for emergency patients (Proposition 10 and Corollary 11).
- **Verified agent limits.** Over a 30-day simulation the agents kept all three bounds of Proposition 12: at most 2 messages per cycle, no message after an opt-out, and report ages under 10 hours for every hospital that replied promptly.

The synthetic instance S1 shows why these results matter. For five of its six anchors, the hospital nearest by distance had no free beds, so a router that looked only at distance would have sent the patient to a full hospital. At anchor A6, the same router would have sent a low-income urgent patient to a private hospital without showing that the nearest affordable public hospital lay 24.2 minutes farther on.

Taken together, the results show that measures for three SDG targets can be estimated for Albay from open data: access to essential destinations by informal transport (target 11.2), access to suitable health care without financial hardship (target 3.8), and the fairness of both across landmarks and LGUs (target 10.2). The estimates still have to be validated in the field (Section 4.3), but they provide an open, verified way to assess spatial access where no baseline transport data existed before.

4.2 Limitations

- **(L1) Headways stand in for schedules.** Informal operators often leave only when the vehicle is full, so their departures are irregular, and the model replaces them with a regular series at an estimated headway. This smooths over real behavior and makes headway the least supported input of the model. The half-headway waiting model and the ±50 percent headway perturbation listed in Table 23 bound the effect, but they do not remove it.

- **(L2) Inferred routes versus driver behavior.** Routes on the road network follow from the vehicle-class limits and from fixed terminal locations. Drivers, however, take shortcuts, switch roads at quiet hours, and change routes on market days, and the model captures none of this. This is the largest difference between the model and actual operations, and it is the main reason why field validation is essential, not optional, for this work (Section 4.3).
- **(L3) Incomplete registers.** Coverage is much better in town centers than in remote areas. The health station of an upland barangay may be missing from every national register, and even when it is listed, OpenStreetMap is unlikely to show the narrow paths up the hill or the bridges and their load limits. The resulting bias overstates access for hard-to-reach communities, and the model has no way to correct it.
- **(L4) No disruption scenarios.** A typhoon, a Mayon Volcano alert, or local flooding can close roads for days, but the feed models only normal weekday and Saturday service.
- **(L5) Imperfect destination data.** Facility records lag behind actual conditions; bed and classroom counts are reported by the facilities themselves and are not independently audited; and new schools or health units may be missing from the data.
- **(L6) Choice of anchor set.** Landmark anchoring trades the modifiable areal unit problem for a selection problem, since adding or removing a single terminal changes the index set and, with it, every inequality statistic downstream. To limit this, the inclusion rules of Appendix C were fixed before listing began, the anchor list is released with the feed, and Section 3.6.1 repeats the analysis on three different anchor sets.
- **(L7) Provincial boundary.** Trips that cross into Sorsogon or Camarines Sur, or go by ferry to Catanduanes, are cut off at the border, which understates access for anchors near the boundary. Hospital mode shares this limit and will never suggest a suitable hospital just across the provincial line.
- **(L8) S1 is synthetic.** Tables 15 to 19 confirm only that the pipeline runs according to the rules of the framework; they contain no empirical findings about Albay. In particular, the Gini coefficient of 0.2747 in Table 17 comes from a synthetic seven-stop network and must not be read as a first estimate for the province.
- **(L9) Hospital reporting is voluntary.** The network can ask, but it cannot verify the counts it receives or compel a reply. An inaccurate count that passes the plausibility checks stays in force until a later report replaces it, and a hospital that does not answer remains UNKNOWN, a safe but uninformative state.

- **(L10) Acuity is not a clinical assessment.** The application does not examine patients and is not a medical device; the acuity level is chosen by the ambulance crew or the user. The crew's clinical judgment always overrides the application, and the parameters K_E and θ_{stab} are placeholders until the provincial health office calibrates them.
- **(L11) Cost ranking is ordinal.** Equation (11) ranks hospitals by expected out-of-pocket burden using only ownership and program flags. It says nothing about final hospital bills, the actual amount of Malasakit Center assistance, or the specific benefits that PhilHealth covers.
- **(L12) Free-flow travel times.** Ambulance travel times assume unobstructed traffic (Assumptions 4 and 8). Congestion, emergency driving, dispatch queues, handover delays at the hospital, and several ambulances competing for one free bed are not modeled.
- **(L13) Simulated e-mail.** Real e-mail runs into bounced messages, spam filters, staff shift changes, and institutional firewalls, none of which the simulation reproduces. The extraction rules were tested on templates written together with those rules, so how well they parse real, unstructured replies is still unknown. Messages that cannot be parsed are passed to a person for review.
- **(L14) Synthetic hospital data in S1.** The hospital figures in Tables 20 to 22 were invented to demonstrate hospital mode and contain no observations of real facilities.

4.3 Challenges in Extending the Results

Regional extension. Taking the framework beyond one province is an operational problem more than a computational one. Proposition 2 and the run times in Tables 13 and 14 show that an ordinary laptop can handle much larger feeds and road networks. The real bottlenecks are keeping the feeds current as operators change their routes and persuading LGUs to look after the public dataset. The natural custodians are the Municipal Planning and Development Offices, which already prepare the Local Public Transport Route Plans (DILG, 2018; DOTr, 2017). As long as the code stays open, routine data updates will not require any change to the underlying mathematical model.

Field validation. Field validation is the obvious next step. The workflow was designed so that field data can enter without disturbing the rest of the pipeline: a ground-surveyed feed can take the place of the generated feed at Stage 3 of Section 3.1, leaving the routing logic of Section 3.2 and all earlier stages as they are. Comparing the generated feed with a surveyed feed over the same anchors would measure, for the first time, what the inference step costs in accuracy, a question that the literature on informal transport has not yet answered.

Putting hospital mode into operation. Moving hospital mode from simulation to live use raises questions of governance and consent. Three conditions must be met before the agents contact any hospital: the Provincial Health Office must be formally informed; the Scientific Review Committee must approve the communication protocol, including the message templates of Appendix F and the data-handling rules; and each facility must opt in explicitly through its public contact channel. A small pilot with consenting hospitals would then measure what simulation cannot, such as response rates, reply delays, and the way replies are actually worded. If the DOH later publishes a machine-readable feed of live bed data, the Registry agent should take that feed first and keep e-mail inquiries as a fallback.

4.4 Applications

Transport planning. Several uses follow directly from this work. Ranking anchor landmarks by how poorly they are served gives local governments a clear order in which to open new routes or extend existing ones. Planners can test a proposed route before granting a franchise simply by rerunning the accessibility surface with the route added. New schools and health facilities can be sited where they add the most reachable capacity for underserved communities, not on population density alone. Daily commuters get fares, boarding points, and alighting points for trips anywhere in the province, and disaster response and evacuation planning gain from taking the limits of each vehicle type into account.

Health services. Hospital mode goes a step further. Families, barangay health workers, and ambulance crews can find the most suitable hospital quickly, with a stated reason for the choice, before a vehicle sets out. During a typhoon or a Mayon Volcano alert, the Provincial Health Office and the Provincial Disaster Risk Reduction and Management Office (PDRRMO) can follow free beds municipality by municipality as the situation unfolds. Mapping the local payment gap also shows where a new Malasakit Center or more public hospital beds would cut travel time the most for low-income families.

Links to SDG targets. Each application corresponds to a specific SDG target, although actual implementation is still some way off. The atlas and the trip planner, with the official fare discounts built in, serve target 11.2 on accessible public transport. Vehicle-class routing supports targets 9.1 and 11.5 on resilient infrastructure and disaster risk reduction. Hospital mode and bed tracking serve targets 3.8 and 3.d on health services and emergency preparedness. The payment-gap map shows where public funds would do the most to improve access to basic services for low-income communities, the concern of target 1.4.

LGUs can use the inequality measures to follow progress on equal access to health care and transport under targets 10.2 and 10.3, something that totals such as road length or vehicle counts cannot show. The same tools can help keep schools within reach of all children (target 4.a), strengthen preparedness for typhoons and volcanic eruptions (target 13.1), and produce open, usable spatial data (target 17.18).

Scalability and replicability. The framework can be carried well beyond Albay. It needs only OpenStreetMap, public facility registers, open population data, and ordinary hardware, so any province, municipality, or region that depends on informal transport could use it. Global monitoring of indicator 11.2.1 still centers on large metropolitan areas, yet many of the places furthest behind on target 11.2 rely on unmapped minibus and jeepney networks. For smaller towns and rural areas, an open-data landmark atlas offers a practical way to close that gap.

The hospital component is light and easy to scale as well. It collects bed counts by ordinary e-mail without any central national platform, so a local health office can give patients and ambulance crews current information quickly, and its payment filter can be adjusted to the health-financing safeguards of other jurisdictions. Field validation is still needed, but the study offers an open and verified framework for assessing accessibility where data are scarce, in keeping with the central promise of the 2030 Agenda to leave no one behind (UNGA, 2015).

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APPENDIX A. OPEN DATA SOURCE REGISTER

Every model input comes from one row of Table A1. None of the rows contains field observations, and the last row (bed-status replies) was not used in any computation reported here. The project repository records access dates and source checksums, so the exact input configuration can always be reproduced.

Table A1. Where each layer of the model comes from.

Source	Publisher	Layer Filled	Access
OpenStreetMap extract for the Bicol Region (OpenStreetMap contributors, 2026)	OpenStreetMap Foundation	Road and walking graphs, road attributes, junction anchors	Open Database License
2020 Census of Population and Housing (PSA, 2021)	PSA	Barangay populations behind the anchor weights	Public release
Philippine Standard Geographic Code (PSA, n.d.)	PSA	LGU and barangay codes; barangay count	Public web service
Land areas of LGUs (PhilAtlas, n.d.)	PhilAtlas, from PSA figures	Land area in Table 1	Public web page
National Health Facility Registry (DOH, n.d.-b)	DOH	Hospitals, infirmaries, and rural health units, with ownership and authorized beds	Public directory
Hospital licensing records under the DOH (2012) classification	Health Facilities and Services Regulatory Bureau, DOH	Service level η_h and emergency-room flag e_h	Public listing
List of accredited hospitals and infirmaries (Philippine Health Insurance Corporation, 2026)	Philippine Health Insurance Corporation	Accreditation flag for the cost class q_h	Public listing
Malasakit Center directory (DOH, n.d.-a)	DOH	Malasakit flag for the cost class q_h	Public listing
Port listings (Philippine Ports Authority, n.d.)	Philippine Ports Authority	Ports and piers used as anchors	Public listing
Contact addresses of facilities	Registry entries and official facility websites	Address m_h for the inquiry of Appendix F	Public, per facility
Masterlist of Schools (Department of Education, 2021)	Department of Education	Public schools (elementary and secondary), with classroom counts where available	Public release

Source	Publisher	Layer Filled	Access
Fare adjustment effective September 28, 2026 (Daguno-Bersamina, 2026; Nieto, 2026)	LTFRB	Fare parameters of Table 4	Public announcement
Route-plan annexes of LGUs, where published (DILG, 2018; DOTr, 2017)	Municipal and city planning offices	Terminal anchors and named corridors	Depends on the LGU
Disaster risk reduction plans of LGUs, where published	Municipal and city DRRM offices	Designated evacuation centers	Depends on the LGU
Bed-status replies (live operation only)	Each participating hospital	Reported capacity, occupancy, and emergency-room status (Definition 14)	Only with the facility's consent; none used here

APPENDIX B. GTFS FILES PRODUCED

Table B1. Contents of the feed and of its extension files.

File	Contents
agency.txt	Operator or association name, service area, contact details, time zone
routes.txt	Route ID, short and long names, mode
trips.txt	One record for each modeled vehicle run
stops.txt	Boarding points inferred by the model, with names and coordinates
stop_times.txt	Arrival and departure time at each stop of each trip
calendar.txt	Service days, with weekday and Saturday patterns kept apart
shapes.txt	Generated route geometry
frequencies.txt	Headway-defined service wherever departure times are not fixed
fare_attributes.txt	Per mode: base fare, base distance, and rate per succeeding kilometer
fare_rules.txt	How fares attach to routes and zones
landmarks.txt (extension)	Anchor and destination inventory, with class, source register, and size w_p
hospitals.txt (extension)	Hospital records of Definition 12: service level, emergency-room flag, ownership, PhilHealth and Malasakit flags, cost class, and contact address
bed_status.txt (extension, live operation only)	Latest validated report of each hospital: time stamp, reported capacity and occupancy, emergency-room status, free beds, and status. It is served by the Status agent and is never part of the published feed.

APPENDIX C. RULES FOR INCLUDING LANDMARKS AND THE INVENTORY SCHEMA

The criteria below were settled before the first landmark was listed. Because the anchor set is a modeling choice, a choice made after seeing the results could not be checked by any reader (Limitation L6).

Inclusion criteria. A place enters the landmark set only if it meets all four of the following conditions.

1. At least one register in Table 2 lists it, under a name familiar to residents.
2. Its published coordinate lies no more than 150 m from a road-network node, so that its entrance node can be assigned mechanically.
3. It is currently operating. Facilities that their registers mark as closed, proposed, or under construction are left out.
4. It has a size if it is a destination. The register supplies w_p; otherwise $w_p = 1$ and the destination is simply counted, and a single type never mixes counted and sized destinations.

Exclusion rules. Private schools go into a separate layer, because the atlas concerns the capacity that government provides. Private hospitals also go into that separate layer for the purposes of the atlas, but hospital mode reads it as well, since patients who can afford private care, and all emergency patients, may be sent to private hospitals. Records of one place found in several registers are merged when their coordinates lie within 50 m of each other and their names are alike, and every merge is logged. A landmark that fails the second criterion is flagged and left out; it is never moved to make it pass.

Table C1. Fields of the landmark inventory, with the extra fields kept for hospitals.

Field	Type	Description
landmark_id	string	Permanent identifier assigned by the project
name	string	Name as written in the source register
class	enum	A class from Table 2
role	enum	anchor, destination, or both
lat, lon	decimal	Coordinate as published (WGS 84)
entrance_node	integer	Road node that serves as o_a or as the destination point
snap_distance_m	decimal	Meters between the published coordinate and the entrance node
size_w	integer	Beds, classrooms, or service windows; 1 for a counted destination

Field	Type	Description
lgu_code	string	Philippine Standard Geographic Code of the LGU that contains the landmark
source	string	Register the record came from
source_date	date	Date on which the register was consulted
merge_log	string	Identifiers of the records merged into this one
Hospitals and infirmaries only		
service_level	integer	η_h: 0 for an infirmary, 1 to 3 for a hospital level (Table 5)
er_flag	boolean	Whether an emergency room is present
ownership	enum	government or private
philhealth_accredited	boolean	Whether the facility is on the PhilHealth accreditation list
malasakit_center	boolean	Whether a Malasakit Center is present
cost_class	integer	q_h by Equation (11)
contact_email	string	Official e-mail address of the facility (never a personal one)

APPENDIX D. APPLICATION FEATURE CHECKLIST

Each feature below ends with the SDG target or targets it is meant to support (UNGA, 2015). The features are planned or under development and have not been tested with outside users.

Transport features

1. Door-to-door journey planning within the study area, telling the user when they will arrive, how much they will pay, which vehicle to take, and where to board and alight (SDG 11.2).
2. A discount switch for senior citizens, students, and persons with disabilities (SDG 11.2 and 10.2).
3. Walking directions for the first and last mile, with a cap on walking distance (SDG 11.2).
4. A choice among the ten vehicle classes of Table 3, giving a road route that the chosen class can use and marking any restricted segments (SDG 9.1 and 11.5).
5. A fare lookup between any two named landmarks (SDG 11.2).
6. A 30-minute landmark accessibility map, with a slider for other budgets (SDG 11.2, 10.3, and 4.a).
7. Offline storage of the road extract and the feed for times of weak signal, such as typhoons (SDG 11.5 and 13.1).
8. A button for reporting errors, so that riders can help correct the dataset once it is released. This is the only way field observations could reach the transport dataset, and only after publication; corrections are logged by date, and none was used in this paper (SDG 17.18).

Hospital mode

9. A hospital-mode switch that leaves only hospitals and infirmaries on the map and changes the guidance to step-by-step directions to a hospital (SDG 3.8).
10. Two travel modes: the ambulance, which uses the ambulance class of Table 3 on the road network, and the patient, who rides public transport and walks.
11. Plain-language choices for the five acuity levels of Table 9, made by the crew or by the user.
12. A choice among the four payment classes of Table 6. The payment filter is switched off automatically at acuity levels 1 and 2, so emergency routing is never delayed by cost (SDG 3.8 and 1.4).

13. A hospital card showing the route, travel time, care level, cost class, and current free beds of the nearest suitable facility, together with the age of the report. A status that is not known is shown as UNKNOWN, never as available (SDG 3.8 and 3.d).

14. A stabilization prompt. For acuity level 1, if an open emergency room is nearer than the nearest suitable hospital by more than θ_{stab}, the application first directs the patient there for stabilization (Definition 18; SDG 3.8).

15. A one-tap button on every hospital-mode screen to call 911, in line with Executive Order No. 56 (2018).

16. A read-only dashboard for the Provincial Health Office showing current free beds and report freshness by municipality (SDG 3.d).

APPENDIX E. RISK REGISTER

The work poses no physical risk to human subjects. The main ethical and operational concerns are the accuracy of the model, transparent reporting, and, once hospital mode goes live, the safety of the people who act on its advice.

Table E1. Risks and how they are handled.

Risk	Likelihood	Mitigation
Model output mistaken for real measurements	High	Every computed value is labeled as a model output. No synthetic result is presented as an empirical finding about a real facility, and the provenance note of Section 3.0 is kept.
Generated routes differ from actual operator routes	High	Acknowledged as Limitation L2. Routes are anchored to published terminals and markets, and comparison with surveyed feeds is deferred to future work.
Missing road attributes in OpenStreetMap	High	Missing attributes are set to +∞ and flagged, and the share of flagged segments is reported with every vehicle-class result.
A hospital does not respond, so its status is unknown	High	The hospital stays in the eligible set as UNKNOWN. The age of its last report is shown, and unknown capacity is never presented as available (Definition 16).
The landmark register is out of date	Moderate	Source and access date are recorded for every entry (Appendix C). Coverage is reported by municipality, and gaps are read as understating spatial disadvantage (Limitation L3).
An unchecked software shortcut produces wrong routes	Moderate	Core routing outputs were checked against independently written reference code before any analysis (Tables 11, 12, and 19).
Inequality measures depend on the choice of anchors	Moderate	Inclusion rules were fixed before listing began (Appendix C). The sensitivity analysis uses three anchor sets (Section 3.6.1), and full anchor lists are released with the dataset.
Typing or parsing errors in bed counts	Moderate	Rules V2 to V4 send unclear replies to human review. All 67 simulated typing errors were caught and held during testing (Table 12).
Hospital staff find the inquiries a nuisance	Moderate	Messages are limited to two per cycle (Proposition 12), a once-a-day option is offered, and opt-out requests take effect before the next send.
The system is used for patient triage	Moderate	The user or crew always selects the acuity level. Clinical judgment overrides the system, every screen has a direct-dial 911 button, and clear on-screen notices state that the application does not assess patients (Limitation L10).

Risk	Likelihood	Mitigation
The fare parameters of Table 4 become outdated	Moderate	Fare parameters are kept in a single dated configuration file, and every downstream result is recalculated by script with no manual data entry.
Cost diverts an emergency patient from the nearest capable facility	Low	Acuity levels 1 and 2 bypass the payment filter automatically (Definition 16; Republic Act No. 10932, 2017). A zero gap was confirmed in 80,000 simulated test cases (Table 12).
A forged or spoofed reply is received	Low	Rule V1 checks the sender and requires a one-time reference token. All 200 attempted forgeries were rejected.
Personal information is received by mistake	Low	Inquiries ask only for aggregate counts. Any reply containing unneeded details is held for human review, and text other than the counts is deleted permanently (Republic Act No. 10173, 2012).
Loss of source code or dataset state	Low	Version control with remote backups. Every reported figure can be traced to a specific script and a verified input checksum.

APPENDIX F. THE INQUIRY MESSAGE AND ITS REPLY FORM

The template used by the inquiry agent is shown below. Words in capitals mark the terms that the Reader looks for, and the subject line carries a one-time reference token (rule V1). This message has never been sent to a real facility; it was written for testing only.

> **Subject:** Request for bed status [BSI-7c2e90d4a1b8]
>
> **To:** the facility's published official address
>
> Good day.
>
> This message was sent automatically by a bed-availability service built as a student research project at Legazpi City Science High School in Legazpi City; our research adviser is Dick Anthony M. Calleja. The service was built so that ambulance crews and patients in Albay can find a hospital with a bed available.
>
> Only three items are needed, and no patient information should be included. Kindly reply to this message, keep the code in the subject line as it is, and complete the three lines below:
>
> AUTHORIZED BED CAPACITY:
>
> BEDS NOW OCCUPIED:
>
> EMERGENCY ROOM STATUS: ACCEPTING / FULL
>
> The app will show your answer as free beds, together with the time you replied. You will get this request at most three times a day, with no more than one reminder after each. To receive it only once a day, reply ONCE A DAY; to stop it, reply REMOVE US.
>
> Thank you very much.
>
> [project e-mail address] | Bitano, Legazpi City | Legazpi City Science High School

Besides the reply form, typed in capitals or in lowercase, the Reader accepts free text in English or Filipino. It recognizes capacity and occupied or free bed counts whenever they are stated plainly, for example "Okupado: 61" or "Kabuuang kama: 85." When a reply gives only a percentage, two different numbers for the same item, or no numbers at all, the Reader does not guess; the reply goes to a human reviewer.

ETHICS STATEMENT

The study is purely computational. No human subjects took part and no personal data were collected. There was no survey or interview, no hospital was contacted, and no chemical, hazardous device, or biological agent was used. No fieldwork was done, and the study proposes none.

Road, transit, landmark, hospital, and population data come either from the public sources of Appendix A or from synthetic instances built for verification and clearly marked as such, for example S1 and the simulated mail server of Section 3.3. All sources are used under their published terms, and OpenStreetMap data carry the attribution that the Open Database License requires (OpenStreetMap contributors, 2026).

Before any real hospital receives a message, the Scientific Review Committee must approve the protocol and the text of Appendix F. Messages will go only to addresses that facilities publish themselves. Every message states that it is automated and part of a student project, names the project adviser, asks only for aggregate bed counts, and asks hospitals to leave out any patient information. In line with Proposition 12, message volume is strictly limited and every opt-out request is honored at once. Bed numbers are shown only per facility and with a time stamp. The project never requests, stores, or displays data on individual patients, and facility addresses are handled as the Data Privacy Act of 2012 (Republic Act No. 10173) requires. Hospital mode only helps with routing. It does not diagnose, triage, or recommend treatment, and it does not replace the clinical judgment of an ambulance crew or of the emergency services (Executive Order No. 56, 2018). The acuity level and payment class that a user enters serve only the current routing request and are then permanently discarded.

Data integrity was a priority throughout the work. No figure in this paper represents a real clinical measurement. Following the provenance rule of Section 3.0, every result obtained from generated or synthetic inputs is labeled as a model output and kept apart from observational data. Plagiarism, data fabrication, and other forms of scientific misconduct were avoided. The hospital-mode and agent-network figures were produced by verify_hospital_mode.py with a fixed random seed, and every figure in the paper can be traced to a specific script, dataset, or test in the project pipeline.

The original research plan had volunteers tracing routes in the field with GPS. The fully computational approach of Section 3.1 replaced that plan, and the change was recorded in the Research Plan Addendum. Because the revision removed human participants from the study, no further review by the Scientific Review Committee was required, and no institutional review board evaluation was needed. Any later field validation or live deployment of the software agents must

first obtain permission from transport operators, local terminal associations, and participating hospitals; must follow established safety protocols for investigators; must leave out personally identifiable information about commuters and patients; and must receive formal ethical clearance before any data are collected.